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**Information technology, Automatic identification and data capture techniques — Bar code
symbol print quality test specification — Two-dimensional symbols**

*Technologies de l'information, Informatique, Techniques d'identification et de saisie de données
automatiques — Spécification de test de qualité d'impression des symboles de code à barres —
Symboles bi-dimensionnels*

Third edition

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Foreword

ISO (the International Organization for Standardization) and IEC (the International Electrotechnical Commission) form the specialized system for worldwide standardization. National bodies that are members of ISO or IEC participate in the development of International Standards through technical committees established by the respective organization to deal with particular fields of technical activity. ISO and IEC technical committees collaborate in fields of mutual interest. Other international organizations, governmental and non-governmental, in liaison with ISO and IEC, also take part in the work. ~~In the field of information technology, ISO and IEC have established a joint technical committee, ISO/IEC JTC 1.~~

~~International Standards~~The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of document should be noted. This document was drafted in accordance with the editorial rules given ~~in~~ of the ISO/IEC Directives, Part 2 (see www.iso.org/directives or www.iec.ch/members_experts/refdocs).

~~The main task of the joint technical committee is to prepare International Standards. Draft International Standards adopted by the joint technical committee are circulated to national bodies for voting. Publication as an International Standard requires approval by at least 75 % of the national bodies casting a vote.~~

~~Attention is drawn~~ISO and IEC draw attention to the possibility that ~~some of the elements~~implementation of this document may ~~be involve~~ the ~~subject~~use of (a) patent(s). ISO and IEC take no position concerning the evidence, validity or applicability of any claimed patent rights in respect thereof. As of the date of publication of this document, ISO and IEC had not received notice of (a) patent(s) which may be required to implement this document. However, implementers are cautioned that this may not represent the latest information, which may be obtained from the patent database available at www.iso.org/patents and <https://patents.iec.ch>. ISO and IEC shall not be held responsible for identifying any or all such patent rights.

~~Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.~~

~~For an explanation of the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT) see www.iso.org/iso/foreword.html. In the IEC, see www.iec.ch/understanding-standards.~~ISO/IEC 15415.

~~This document~~ was prepared by Joint Technical Committee ISO/IEC JTC 1, *Information technology*, Subcommittee SC 31, *Automatic identification and data capture techniques*.

This third edition cancels and replaces the second edition (ISO/IEC 15415:2011) which has been technically revised.

The main changes ~~compared to the previous edition~~ are as follows:

- ~~introduction of —~~ a continuous (or decimal) grading ~~has been introduced~~;
- ~~introduction of —~~ a more optimal threshold calculation ~~has been introduced~~;
- ~~introduction of —~~ a more stable symbol contrast calculation ~~has been introduced~~;
- ~~—~~ a definition of grading for print growth ~~has been added~~;

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- combination of modulation and reflectance margin into a single measurement.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html and www.iec.ch/national-committees.

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Introduction

The technology of bar coding is based on the recognition of patterns encoded, in bars and spaces or in a matrix of modules of defined dimensions, according to rules defining the translation of characters into such patterns, known as the symbology specification. Symbology specifications may be categorised into those for linear symbols, on the one hand, and two-dimensional symbols on the other; the latter may in turn be ~~sub-divided~~~~subdivided~~ into “multi-row bar code symbols” sometimes referred to as “stacked bar code symbols”, and “two-dimensional matrix symbols”. In addition, there is a hybrid group of symbologies known as “composite symbologies”; these symbols consist of two components carrying a single message or related data, one of which is usually a linear symbol and the other a two-dimensional symbol positioned in a defined relationship with the linear symbol.

Multi-row bar code symbols are constructed graphically as a series of rows of symbol characters, representing data and overhead components, placed in a defined vertical arrangement to form a (normally) rectangular symbol, which contains a single data message. Each symbol character has the characteristics of a linear bar code symbol character and each row has those of a linear bar code symbol; each row, therefore, may be read by linear symbol scanning techniques, but the data from all the rows in the symbol must be read before the message can be transferred to the application software.

Two-dimensional matrix symbols are normally square or rectangular arrangements of dark and light modules, the centres of which are placed at the intersections of a grid of two (sometimes more) axes; the coordinates of each module need to be known in order to determine its significance, and the symbol must therefore be analysed two-dimensionally before it can be decoded. Some matrix codes are comprised of unconnected dots, in which the individual modules do not directly touch their neighbours but are separated from them by a clear space.

Unless ~~the context requires~~~~explicitly specified~~ otherwise, the term “symbol” in this document ~~may refer~~~~refers~~ to either type of symbology.

The bar code symbol must be produced in such a way as to be reliably decoded at the point of use, if it is to fulfil its basic objective as a machine-readable data carrier.

Manufacturers of bar code equipment and the producers and users of bar code symbols therefore require publicly available standard test specifications for the objective assessment of the quality of bar code symbols ~~for~~~~a process known as verification~~, to which they can refer when developing equipment and application specifications or determining the quality of the symbols. Such test specifications form the basis for the development of measuring equipment for process control and quality assurance purposes during symbol production as well as afterwards.

The performance of measuring equipment for the verification of symbols ~~is~~~~also known as verifiers~~, is ~~the subject of a separate document (covered in~~ ISO/IEC 15426, ~~Parts 1 and~~ ISO/IEC 15426-2).

The methodology described in this document is intended to achieve comparable results to the linear bar code symbol quality standard ISO/IEC 15416, the general principles of which ~~it~~~~this document~~ has followed. It should be read in conjunction with the symbology specification applicable to the bar code symbol being tested, which provides symbology-specific ~~detail~~~~details~~ necessary for its application. Two-dimensional multi-row bar code symbols are verified according to the ISO/IEC 15416 methodology, with the modifications described in Clause 6; different parameters and methodologies are applicable to two-dimensional matrix symbols. The procedures described in this document must necessarily be augmented by the reference decode algorithm and other measurement details within the applicable symbology specification, and they may also be altered or overridden as appropriate by governing symbology or application specifications.

The method of quality assessment described in this document is most applicable to reading environments wherein printed and otherwise marked symbols are read predominantly by diffuse reflection. For direct part mark applications, in which symbols and substrates may be glossy, specular, low contrast, etc., a modified

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and extended version of the methodology defined in this document has been defined in ISO/IEC 29158 and provides for lighting arrangements and enhanced algorithms which more closely match reading equipment commonly used in DPM applications.

NOTE The Bibliography provides official and industry standards containing symbology specifications, among other references, to which this document applies. However, the Bibliography does not provide an exhaustive list of symbology specifications.

Information technology, Automatic identification and data capture techniques — Bar code symbol print quality test specification — Two-dimensional symbols

1 Scope

This document

- specifies two methodologies for the measurement of specific attributes of two-dimensional bar code symbols; ~~one of these being applicable~~which applies to multi-row bar code symbologies and the other to two-dimensional matrix symbologies;
- ~~defines~~specifies methods for evaluating and grading these measurements and deriving an overall assessment of symbol quality;
- gives information on possible causes of deviation from optimum grades to assist users in taking appropriate corrective action.

This document applies to ~~those~~ two-dimensional symbologies for which a reference decode algorithm has been defined, ~~but it is~~however the methodologies in this document can be applied partially or wholly to other similar symbologies.

NOTE While this document ~~may~~can be applied to direct part marks, ~~it is possible that~~ better correlation between measurement results and scanning performance ~~will~~can be obtained with ISO/IEC 29158 in combination with this document.

2 Normative references

The following ~~referenced~~ documents are ~~indispensable for~~referred to in the ~~application~~text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 19762, *Information technology — Automatic identification and data capture (AIDC) techniques — Harmonized vocabulary*

ISO/IEC 15416, *Information technology – Automatic identification and data capture techniques — Bar code print quality test specification — Linear symbols*

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ISO/IEC 15426-2, *Information technology — Automatic identification and data capture techniques — Bar code verifier conformance specification — Part 2: Two-dimensional symbols*

NOTE — The Bibliography lists official and industry standards containing specifications of symbologies to which (inter alia) this document is applicable.

3 Terms and definitions

For the ~~purpose~~~~purposes~~ of this document, the terms and definitions given in ISO/IEC 19762, ISO/IEC 15416 and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at <https://www.iso.org/obp>

— IEC Electropedia: available at <https://www.electropedia.org/>

3.1

binarised image

binary (black/white) image created by applying a threshold to the *pixel* ~~(3.7)~~ values in the *reference grey-scale image*: ~~(3.9)~~

3.2

effective resolution

resolution obtained on the surface of the symbol under test, ~~normally expressed in pixels per mm or pixels per inch~~, and calculated as the resolution of the image capture element multiplied by the magnification of the optical elements of the measuring device.

Note 1 to entry: The effective resolution is normally expressed in *pixels* (3.7) per mm (or pixels per inch).

3.3

error correction capacity

number of codewords, or units of data, in a symbol (or error control block) assigned for erasure and error correction, minus the amount reserved for error detection.

3.4

inspection area

portion of an image which contains the entire symbol to be tested inclusive of its quiet zones.

3.5

grade threshold

boundary value separating two grade levels, the value itself being taken as the lower limit of the upper grade.

3.6

module error

module of which the apparent dark or light state in the *binarised image* ~~(3.1)~~ is inverted from its intended state.

3.7

pixel

individual light-sensitive element in an array, ~~e.g.,~~

EXAMPLE — CCD (charge coupled device) ~~or~~, CMOS (complementary metal oxide semiconductor) device.

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3.8
raw image

plot of the reflectance values in x and y coordinates across a two-dimensional image, representing the discrete reflectance values from each *pixel* (3.7) of the light-sensitive array.

3.9
reference grey-scale image

plot of the reflectance values in x and y coordinates across a two-dimensional image, derived from the discrete reflectance values of each *pixel* (3.7) of the light-sensitive array by convolving the *raw image* (3.8) with a *synthesised aperture* (3.10).

3.10
synthesised aperture
convolutional kernel used to blur an image

4 Symbols {and abbreviated terms}

4.1 Symbols

The following symbols are used in this document:

4.1.1 ~~E_{cap}~~

Error correction capacity of the symbol

4.1.2 ~~D~~

Distance across, or with of an element in a symbol

4.1.3 ~~D_{NOM}~~

Expected, or nominal, width of an element in a symbol

4.1.4 ~~e~~

number of erasures

4.1.5 ~~M_{MOD}~~

a measure of the difference in reflectance between a module and the threshold

4.1.6 ~~M_{ANU}~~

A measure of axial nonuniformity

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4.1.7 ~~M_{GNU}~~

A measure of grid nonuniformity

4.1.8 ~~M_{UEC}~~

A measure of unused error correction taking into account the number of errors, erasures and capacity

4.1.9 ~~R~~

reflectance

4.1.10 ~~$R_{\max}$~~

reflectance value indicative of the brightest area of a symbol

4.1.11 ~~$R_{\min}$~~

reflectance value indicative of the darkest area of a symbol

4.1.12 ~~ΔR_{Sc}~~

Symbol contrast, the difference in reflectivity between $R_{\max}$ and $R_{\min}$

4.1.13 ~~t~~

number of errors

4.1.14 ~~T~~

Threshold

4.1.15 ~~X_{AVG}~~

The average spacing of modules in the horizontal axis

4.1.16 ~~Y_{AVG}~~

The average spacing of modules in the vertical axis

E_{cap} error correction capacity of the symbol

D distance across or width of an element in a symbol

D_{NOM} expected or nominal width of an element in a symbol

e number of erasures

M_{MOD} measure of the difference in reflectance between a module and the threshold

M_{ANU} measure of axial nonuniformity

M_{GNU} measure of grid nonuniformity

M_{UEC} measure of unused error correction, accounting for the number of errors, erasures and capacity

R reflectance

$R_{\max}$ indicative reflectance of the brightest area of a symbol

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R_{min}	<u>indicative reflectance of the darkest area of a symbol</u>
ΔR_{SC}	<u>symbol contrast, i.e. the difference in reflectance between R_{max} and R_{min}</u>
t	<u>number of errors</u>
T	<u>threshold</u>
X_{AVG}	<u>average spacing of modules in the horizontal axis</u>
Y_{AVG}	<u>average spacing of modules in the vertical axis</u>

4.2 Abbreviated terms

4.2.1 ANU

~~1.1.1 Axial nonuniformity~~

4.2.2 FPD

~~Fixed Pattern Damage~~

4.2.3 GNU

~~grid nonuniformity~~

4.2.4 MOD

~~modulation~~

4.2.5 PG

~~Print growth~~

4.2.6 SC

~~Symbol contrast~~

4.2.7 UEC

~~Unused error correction~~

<u>ANU</u>	<u>axial nonuniformity</u>
<u>CU</u>	<u>contrast uniformity</u>
<u>FPD</u>	<u>fixed pattern damage</u>
<u>GNU</u>	<u>grid nonuniformity</u>
<u>MOD</u>	<u>modulation</u>
<u>PG</u>	<u>print growth</u>
<u>SC</u>	<u>symbol contrast</u>

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UEC unused error correction

5 Quality grading

5.1 General

The measurement of two-dimensional bar code symbols is designed to yield a quality grade, reported as ~~Symbol Grade~~the symbol grade, indicating the overall quality of the symbol. This can be used by producers and users of the symbol for diagnostic and process control purposes and is broadly predictive of the read performance to be expected of the symbol. The process requires the measurement and grading of defined parameters, from which the ~~Symbol Grade~~symbol grade is derived. The ~~Symbol Grade~~symbol grade is sometimes called the “overall symbol grade” when used in a context in which the modifier “overall” helps to clarify and distinguish the symbol grade from other parameter grades.

As a consequence of the use of different types of reading equipment under differing conditions in actual applications, the levels of quality required of two-dimensional bar code symbols to ensure an acceptable level of performance will differ between different real-world applications. Application specifications should therefore define the required symbol quality in terms of aperture, light (angle, orientation, and wavelength), and minimum symbol grade in accordance with this document. The guidelines in Annex C are provided as an aid in writing application specifications ~~which employ~~according to this document. Annex D provides additional information regarding substrate properties. When this document is used without an applicable application specification, suitable choices for aperture, light (angle and wavelength), and minimum grade, must be made by the user. However, if these choices are not suited to the intended application, then the achieved symbol grade cannot be relied upon as an indicator of reading performance in the intended application. Thus, it is highly preferred to follow an agreed upon application specification when using this document (such as a specification mutually agreed upon or published by an industry standards body or regulatory authority).

This document defines the method of obtaining a quality grade for individual symbols. The use of this method in high volume quality control regimes may require sampling in order to achieve desired results. Such sampling plans, including required sampling rates are outside of the scope of this document.

NOTE Information on sampling plans ~~may can~~ be found in ~~the following~~ ISO 3951-1, ISO 3951-2, ISO 3951-3, ISO 3951-5 ~~or and~~ ISO 28590.

5.2 Expression of quality grades

This document specifies a numeric basis for expressing quality grades on a descending scale from 4,0 to 0 in steps of 0,1. The highest quality is represented by 4,0. ~~Individual parameter grades and overall symbol grades may also be expressed informatively, and with a loss of precision, on a corresponding alphabetic scale from A to F, which is sometimes used in application specifications for historical reasons. Alphabetic grades are not formal and should not be reported as the formal grade.~~ The link between numeric and alphabetic grades is given in Table 1. Numeric grades are more precise because they express a grade with ten steps within each letter grade range. Overall numeric grades should not be rounded to the nearest whole number. When using alphabetic grades for convenience they shall be accompanied by a numerical grade. ~~Alphabetic grades are not normative and should not be reported as the formal grade.~~

Table 1 ~~mapsshow~~s the ~~correspondence between~~ alphabetic and numeric grades ~~to each other~~.

Table 1 — Correspondence between numeric and historical alphabetic quality grades

Numeric range	Alphabetic grade
≥3,5 (i.e., 3,5; 3,6; 3,7; 3,8; 3,9; 4,0)	A

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2,5 to 3,4	B
1,5 to 2,4	C
0,5 to 1,4	D
≤0,4 (i.e. 0,0; 0,1; 0,2; 0,3; 0,4)	F

5.3 Symbol grade

The symbol grade shall be calculated ~~as defined in~~ accordance with 6.2.6 or 6.3 or 7.6 or Clause 8 and as applicable to the type of symbol.

5.4 Specifying the symbol grade requirement in an application specification

~~An Application Specification must~~An application specification shall specify the minimum grade requirement as well as the grading conditions, shown in the format:

- grade/aperture/light/angle;
- where:
- ~~"grade" is the minimum grade required;~~
 - ~~"aperture" is the aperture reference number (from ISO/IEC 15416 for linear scanning techniques), or the diameter in thousandths of an inch (to the nearest thousandth) of the aperture defined in 5.6.2;~~
 - ~~"light" defines the illumination: a numeric value indicates the peak light wavelength in nanometres (for narrow band illumination); the alphabetic character W indicates broadband (nominally "white light"), the spectral response characteristics of which must imperatively be defined or specified in kelvin in parenthesis after the W designated by a number followed by the letter K, or have their source specification clearly referenced;~~
 - ~~"angle" is an additional parameter specifying the angle of incidence (in degrees, relative to the plane of the symbol) of the illumination from four sides (as default, unless another orientation is specified by an application), see Figure 1 and Figure 2. It~~
- grade indicates the minimum grade required;
 - aperture indicates the aperture reference number (from ISO/IEC 15416 for linear scanning techniques), or the diameter in thousandths of an inch (to the nearest thousandth) of the aperture defined in 5.6.2;
 - light indicates the numeric value of the peak light wavelength (i.e. the illumination) in nanometres (for narrow band illumination); the alphabetic character W indicates broadband (nominally, "white light"), the spectral response characteristics of which must imperatively be defined or specified in kelvin in parenthesis after the W designated by a number followed by the letter K, or have their source specification clearly referenced;
 - angle indicates the angle of incidence (an additional parameter) in degrees (relative to the plane of the symbol) of the illumination from four sides (as a default, unless another orientation is specified by an application) – see Figure 1 and Figure 2.

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The angle shall be included in the specification of grading requirements when the angle of incidence is other than 45°. While it may be included for 45°, its absence indicates that the specified angle of incidence is 45° by default. Application specifications may specify a different angle of incidence instead of leaving it blank.

The orientation of light refers to the number of directions, the default being four. Light from only two sides may be specified by appending the letter "T" to the angle, such as for example 30T. The letter "S" may be used to indicate only one side. The letter "Q" may be used to indicate four sides, but omitting it implies four sides so 45Q is equivalent to 45 but 45Q is explicit regarding the orientation.

For example, an **EXAMPLE 1** An application specification may specify a symbol quality requirement as 1,5/05/660 to indicate that the required symbol quality is 1,5 or higher obtained using an aperture size of 0,125 mm (reference number 05) with 660 nm light incident from 45 degrees from four sides.

Another example, using **EXAMPLE 2** Using a white light with 5400 K, the light source would be 1,5/05/W(5400 K).

NOTE—Other lighting options are defined in ISO/IEC 29158 which may be more appropriate for direct part marking applications, especially in applications which utilize symbols marked on reflective substrates. Therefore, ISO/IEC 29158 should be specified by an application specification as the method of grading when such lighting options are preferred in an application.

5.5 Reporting of symbol grade

A symbol grade is only meaningful if it is reported in conjunction with the illumination and aperture used. It should be shown in the format grade/aperture/light/angle, where:

- "grade" is the symbol grade as defined in 5.3;
- "/aperture" is the aperture reference number (from ISO/IEC 15416 for linear scanning techniques, or the diameter in thousandths of an inch (expressed to the nearest thousandth or a to higher precision) of the aperture defined in 5.6.2;
- "/light" defines the illumination: a numeric value indicates the peak light wavelength in nanometres (for narrow band illumination); the alphabetic character W indicates that the symbol has been measured with broadband illumination ("white light") the spectral response characteristics of which must imperatively be defined or have their source specification clearly referenced;

— "/angle

where is an additional parameter defining the angle of incidence (relative to the plane of the symbol) of the illumination from four sides. It

- grade indicates the symbol grade as defined in 5.3;
- aperture indicates the aperture reference number (from ISO/IEC 15416 for linear scanning techniques, or the diameter in thousandths of an inch (expressed to the nearest thousandth or a to higher precision) of the aperture defined in 5.6.2;
- light indicates the numeric value indicates the peak light wavelength (i.e. the illumination) in nanometres (for narrow band illumination); the alphabetic character W indicates that the symbol has been measured with broadband illumination ("white light") the spectral response characteristics of which must imperatively be defined or have their source specification clearly referenced;
- angle indicates the angle of incidence (an additional parameter, relative to the plane of the symbol) of the illumination from four sides.

The angle shall be included in the reporting of the overall symbol grade when the angle of incidence differs from the default orientation of 45° from four sides. While it may be included for the default orientation, its absence indicates that the angle of incidence is 45° from four sides.

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The notation used to specify a minimum grade that is required in an application is similar to the notation used to report a grade result, but the grading requirement specifies a minimum for the grade in an application, whereas the result specifies a specific occurrence of a grade produced by verifying a symbol.

Examples

EXAMPLE 1 2,8/05/660 indicates that the overall grade was 2,8 and was obtained with the use of a 0,125 mm aperture (reference number 05) and a 660 nm light source, incident at 45° from four sides.

EXAMPLE 2 2,8/10/W/30Q indicates that the grade of a symbol was 2,8 and was obtained with broadband light, measured with light incident at 30° from four sides and using a 0,250 mm aperture (reference number 10), but would need to be accompanied either by a reference to the application specification defining the reference spectral characteristics used for measurement or a definition of the spectral characteristics themselves.

EXAMPLE 3 2,8/10/670 indicates that the grade of a symbol was 2,8 and was obtained using a 0,250 mm aperture (reference number 10), and a 670 nm light source, incident at 45° from four sides.

NOTE The previous ~~version~~ ~~edition~~ of this document defined the use of a * to indicate the presence of extreme reflectance in or near the symbol. This has been removed in this ~~revision~~ ~~edition of this document~~.

5.6 Optical ~~Setup~~ ~~setup~~ and ~~Obtaining~~ ~~obtaining~~ the ~~Test Image~~ ~~test images~~

5.6.1 General requirements

Equipment for assessing the quality of symbols in accordance with this clause shall comprise a means of measuring and analysing the variations in the reflectivity of a symbol on its substrate over an inspection area which shall cover the full height and width of the symbol including all quiet zones. All measurements on a two-dimensional matrix symbol shall be made within the inspection area defined in accordance with 5.6.4.

The measured reflectance values shall be expressed in percentage terms by means of calibration to a reference reflectance standard traceable to National Measurement Institutes.

NOTE Maximum white diffuse reflectance is taken as 100 %~~%~~.

The peak light wavelength or, in the case of applications designed for the use of broadband illumination, the reference spectral response characteristics, should be specified in the application specification to suit the intended scanning environment. Light sources may either have inherently narrow band or near-monochromatic characteristics or have broad bandwidths. Special care is necessary when making measurements with broadband illumination. The overall spectral response of the measurement and reading systems shall be defined and matched in order to make accurate and repeatable measurements of the grey-scale reflectance of a sample area that correlate with the intended system. Overall spectral response includes the spectral distribution of the light source, the response of the detector and any associated filter characteristics.

Refer to Annex C for guidance on the selection of the light source.

5.6.2 Convolution with the measuring aperture

The measuring aperture is specified by the application specification to suit the *X* dimensions of the symbols in the application and the intended scanning environment. The measuring aperture is applied by convolution

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the raw image with ~~an aperture, which is typically a~~ synthetic aperture and defined by a kernel of equal weights within a round area and zero outside. The purpose of applying an aperture is

- a) ~~to (1)~~ reduce the effect of small imperfections in the symbol, such as substrate texture and printing defects, and ~~(2)~~
- b) ~~to~~ standardize the operation of verification devices, independent of the pixel density (resolution) of each device.

NOTE The aperture ~~may~~ can be a physical one, as ~~was~~ is common for implementations of ISO/IEC 15416. In any case, this convolving aperture is not to be confused with an aperture of a lens.

5.6.3 Geometry of the optical setup

A reference optical geometry is defined for reflectivity measurements and consists of:

- flood incident illumination, uniform across the inspection area, from a set of four light sources arranged at ~~90 degree~~ intervals of ~~90°~~ around a circle concentric with the inspection area and in a plane parallel to that of the inspection area, at a height which ~~will allow~~ allows incident light to fall on the centre of the inspection area at an angle of 45° (or another angle if specified for the application) to its plane, and
- a light collection device, the optical axis of which is perpendicular to the inspection area and passes through its centre, and which focuses an image of the test symbol on a light-sensitive array.

The light reflected from the inspection area (see 5.6.4) shall be collected and focussed on the light-sensitive array.

Two principles govern the design of the optical set-up.

— First, the test image's grey-scale shall be nominally linear and not be enhanced in any way (such as a gamma correction). Filters which modify the spectral characteristics of the light source should only be used to remove unwanted ambient light and/or to produce the spectral characteristics of illumination specified for an application (such as a particular narrow band). Other filters should be avoided or used only when they are consistent with an application specification (see C.1.1) in which case it shall be reported along with the grade. A filter that merely restricts the spectral components to the specified wavelength need not be reported, but one which introduces special effects, such as a UV (ultraviolet) filter, shall be reported. In any case where the use of a filter significantly affects a measurement, the usage of the filter shall be reported along with the grade.

— Second, the image resolution shall be adequate to produce consistent readings. Implementations should have sufficient resolution irrespective of the rotation, unless the manufacturer defines handling instructions which restricts the angle of the symbol in relation to the camera sensor orientation.

NOTE 1 The conformance requirements in ISO/IEC 15426-2 are useful for confirming adequate resolution.

Alternative optical arrangements may be used in an implementation, provided that the measurements obtained with them can be correlated with the use of the reference or the specified optical arrangement. In particular, the tolerances specified in ISO/IEC 15426-2 shall be met. Furthermore, ~~Application Specifications may define application specifications can establish~~ alternative optical arrangements, in accordance with 5.4, which are better aligned to an application and ~~may thus can~~ result in different results than the reference optical arrangement.

Figures 1 and 2 illustrate the principle of the optical arrangement but are not intended to represent actual devices; ~~in particular~~ the magnification of the device ~~in particular~~ is likely to differ from 1:1.

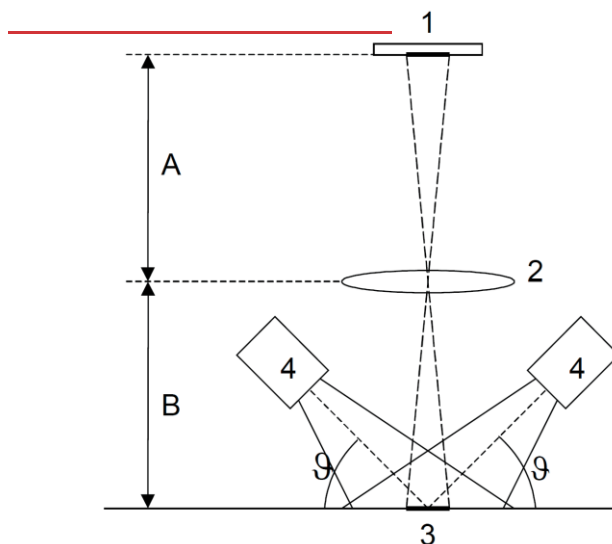
ISO/IEC DIS 15415:2023(E)2024(en)

The general test set-up defined in ~~these clauses~~ this optical geometry should work suitably for many open applications with methodology described in this document. However, for ~~Direct Part Mark~~ direct part mark applications, application specifications should consider using ISO/IEC 29158 instead.

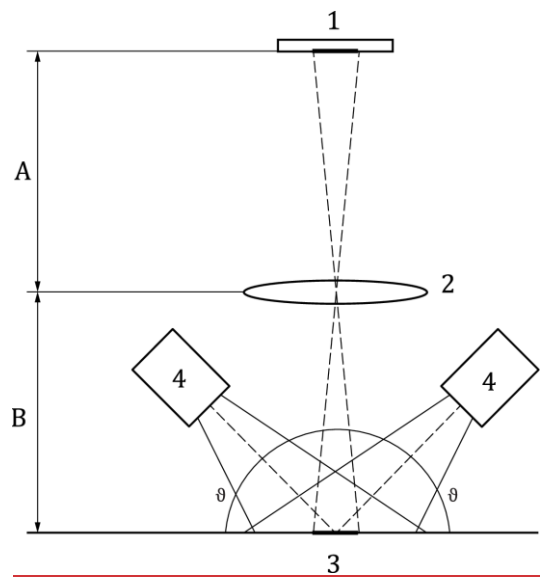
NOTE 2 ISO/IEC 29158 ~~formally defines~~ establishes an extension of the methodology described in this document and is more appropriate for many direct part marking applications.

This reference geometry is intended to provide a basis to assist the consistency of measurement and ~~may does~~ not necessarily correspond with the optical geometry of individual scanning systems. As stated in 5.4, specialised applications, and especially those involving direct part marking which employs physical changes to the surface of the substrate for the creation of the graphic image, ~~may can~~ require the angle of illumination, in particular, to be set to a different particular angle, such as 30° , to the plane of the symbol. If an angle and/or orientation, other than the default is used, then the angle of incidence and orientation of the light shall be stated as a fourth parameter when reporting the overall symbol grade, as described in 5.5.

The modified methodology ~~defined that is specified~~ in ISO/IEC 29158 intended for direct part marking applications defines more illumination options; and ~~it~~ should be used in applications where diffuse lighting from four sides does not provide adequate correlation with the requirements of a specialised application.



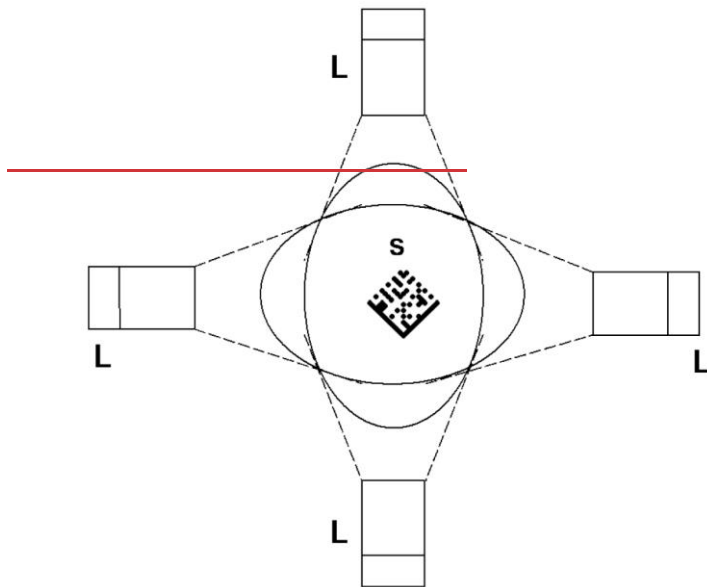
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- Key
- 1 light sensing element
 - 2 lens providing 1:1 magnification (i.e. measurement A = equals to measurement B)
 - 3 inspection area
 - 4 light sources
 - ϑ angle of incidence of light relative to plane of symbol (the default =is equal to 45°, optionally 30° or 90° diffuse)

Figure 1 — Reference optical arrangement ~~—side—~~ Side view

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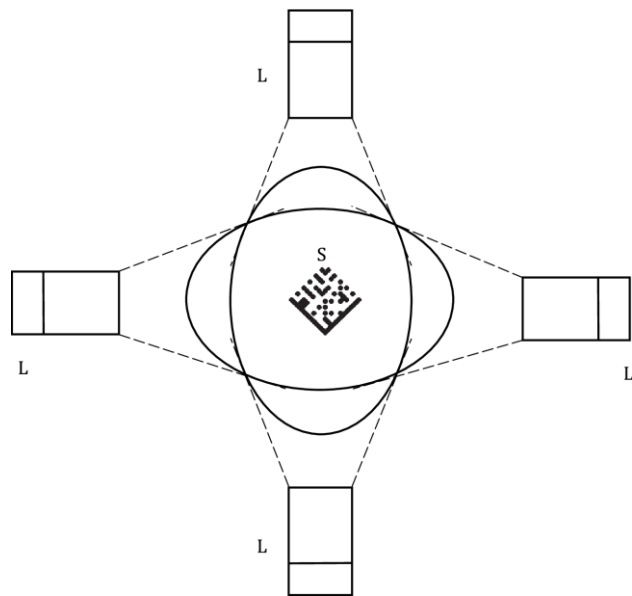
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Key
L light source
S symbol

Figure 2 — Reference optical arrangement ~~—plan—~~ Plan view

5.6.4 Inspection area

The inspection area for all measurements shall be an area framing the complete symbol, including quiet zones. The centre of the inspection area shall be as close as practicable to the centre of the field of view.

NOTE The inspection area is not the same as the field of view of the verifier, which is often larger.

5.6.5 Measurement conditions

A test image of the symbol shall be obtained in a configuration that mimics the typical scanning situation for that symbol, but with substantially higher resolution (see 5.6), uniform illumination, and at best focus. The reference optical arrangement is defined in 5.6.3. Alternative optical arrangements are also discussed in 5.6.3.

Measurements shall be made with the light of a single peak wavelength or a set of spectral characteristics, in accordance with 5.4. Ambient light levels shall be controlled to have no significant influence on the measurement results.

Whenever possible, measurements shall be made on the symbol in its final configuration, i.e. the configuration in which it is intended to be scanned. The geometry of the optical setup described in 5.6.3 is intended to prevent extreme reflectance values outside the symbol area (e.g. when surrounded by free air or a highly specular reflective surface) from distorting the symbol contrast measurements.

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6 Measurement methodology for two-dimensional multi-row bar code symbols

6.1 General

The evaluation of two-dimensional multi-row bar code symbols shall be based on the application of the methodology of ISO/IEC 15416, modified as described in 6.2.2 or 6.3, and if appropriate for the symbology, on the application of the additional provisions described in 6.2.3, 6.2.4 and 6.2.5, to derive an overall symbol grade. Ambient light levels shall be controlled in order not to have any influence on the measurement results. The symbol shall be scanned using the light wavelength(s) and the aperture size specified in the appropriate application specification.

When performing a measurement, the scan lines should

- ~~— be made~~ perpendicular to the height of the bars in the start and stop characters, and ~~should~~
- ~~— pass~~, as far as possible ~~pass~~, through the centres of rows in order to minimise the effect of cross-talk from adjacent rows.

In the case of area imaging techniques, a number of scan lines, perpendicular to the height of the bars and sufficient to cover all rows of the symbol, shall be synthesised to produce scan reflectance profiles as described in ISO/IEC 15416.

6.2 Symbologies with cross-row scanning ability

6.2.1 Basis of grading

The distinguishing feature of these symbologies is their ability to be read with scan lines that cross row boundaries. Symbologies of this type, typically also share the feature that the start and stop patterns (or equivalent features of the symbol, e.g. the ~~Row-Address-Patterns~~ row address patterns of MicroPDF417 symbology specified in ISO/IEC 24728) are constant from row to row, or the position of only one edge in these patterns varies by no more than 1X in adjacent rows of the symbol. These symbologies shall be graded ~~in with~~ respect ~~of to~~

- ~~— Analysis~~ the analysis of the scan reflectance profile (~~based on ISO/IEC 15416~~) (see 6.2.2),
- ~~— Codeword Yield~~ the codeword yield (see 6.2.3),
- ~~— Unused Error Correction~~ the unused error correction (see 6.2.4),
- ~~— Codeword~~ the codeword print quality (see 6.2.5), and
- ~~— Scan~~ the scan reflection profiles that are produced from the raw image by convolving with the specified aperture.

6.2.2 Grade based on analysis of scan reflectance profile

The start and stop or equivalent (e.g. ~~Row-Address~~ row address) patterns of the symbol shall be evaluated ~~according to in accordance with~~ ISO/IEC 15416. Regions with data content ~~will be~~ are evaluated separately as described in 6.2.3, 6.2.4 and 6.2.5. Test scans of the ~~Start~~ start and ~~Stop~~ stop patterns shall be graded using

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all parameters specified in ISO/IEC 15416. The effective aperture size is specified in the appropriate application specification or is the default aperture size appropriate for the symbol X dimension given in ISO/IEC 15416.

For the analysis of the scan reflectance profiles, the number of scans should be ten; or the height of the symbol divided by the measuring aperture if this quotient is less than ten. Scans should be approximately evenly spaced over the height of the symbol. For example, in a twenty-row symbol, the ten scans ~~might~~can be performed in alternate rows. In a two-row symbol, up to five scans ~~might~~can be performed in each row, at different positions in the height of the bars. The symbology specification ~~may~~can give more specific guidance on the selection of the scans to be used.

The methods given in ISO/IEC 15416 shall be used to identify bars and spaces and the edge locations. For the evaluation of the parameters 'decode' and 'decodability' the reference decode algorithm for the symbology shall be applied. Each scan shall be graded as the lowest grade for any individual parameter in that scan. The grade based on scan reflectance profiles shall be the arithmetic mean of the grades for the individual scans, ~~just~~ as defined in ISO/IEC 15416.

The measurement of print growth or loss may be used for process control purposes. ~~Note that~~However, this method ~~will~~is not be sensitive to printing variations parallel to the height of the start and stop characters. If a full analysis of the printing process is desired, symbols should be printed and tested in both orientations.

6.2.3 Grade based on ~~Codeword Yield~~codeword yield

6.2.3.1 ~~This parameter~~Overview of method

Grade based on codeword yield measures the efficiency with which linear scans can recover data from a two-dimensional multi-row symbol. The ~~Codeword Yield~~codeword yield is the number of validly decoded codewords expressed as a percentage of the maximum number of codewords that ~~could have been~~can be decoded (after adjusting for tilt). A poor ~~Codeword Yield~~codeword yield, for a symbol whose other measurements are good, ~~may~~can indicate a Y-axis print quality problem (such as those shown in Table B.1).

6.2.3.2 Qualifying scans

Obtain a matrix of the correct symbol character values, such as would result from successful completion of the ~~UEC~~unused error correction (UEC) calculations (see 6.2.4). This matrix is used as the "final decode of the symbol" used in subsequent steps to determine validly decoded codewords.

An individual scan qualifies for inclusion in the ~~Codeword Yield~~codeword yield calculation if it meets either of ~~the following~~ two conditions:

~~1a)~~ The scan did not include recognised portions of either the top or the bottom row of the symbol. At least one of the ~~Start~~start or ~~Stop~~stop (or ~~Row Address~~row address) patterns shall have been successfully decoded from that scan, together with at least one additional codeword or the corresponding second ~~Start~~start or ~~Stop~~stop pattern, or ~~Row Address Pattern~~row address pattern.

~~2b)~~ The scan included recognised portions of either the top or the bottom row of the symbol. Both the ~~Start~~start and ~~Stop~~stop patterns of the symbol shall have been successfully decoded from that scan.

It is important to note that an extension to the symbology's ~~Reference Decode Algorithm~~reference decode algorithm is required, in order to detect and decode a pair of ~~Start~~start and ~~Stop~~stop patterns when neither of the adjacent codewords is decodable. As examples, a linear search for a matching pair of PDF417 ~~Start~~start and ~~Stop~~stop patterns, or a linear search for a matching pair of MicroPDF417 ~~Row Indicator Patterns~~row indicator patterns, would fulfil this requirement for scans where the ~~Reference Decode Algorithm~~reference decode algorithm alone did not decode both patterns; thus, this extension can qualify a scan where no codewords ~~(other than the matched end patterns)~~ were decoded. ~~Note however, that a~~

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NOTE A scan that contains only a single decoded ~~Start~~**start** or ~~Stop~~**stop** pattern found by this linear search does not count as a qualified scan, if no other codewords or corresponding second ~~Start~~**start** or ~~Stop~~**stop** pattern, or ~~Row Address Pattern, were~~**row address pattern, are** also decoded.

6.2.3.3 DecodeDecoding the symbol completely and ~~populate~~populating** the symbol matrix.**

For each qualified scan, compare the codewords actually decoded with the codewords in the symbol matrix and count the number of codewords that match. Accumulate the total number of validly decoded codewords, and update a count of the number of times each codeword of the symbol has been decoded and a count of the number of times each row has been detected. ~~Also record~~**Record** a count of the number of detected row crossings in each scan (a crossing is “detected” when a scan line yields correctly-decoded codewords from adjacent rows).

6.2.3.4 Calculation and grading

After processing each scan, calculate the maximum number of codewords that ~~could~~**can** have been decoded thus far, as the number of qualified scans multiplied by the number of columns in the symbol (excluding the fixed patterns, such as the ~~Start~~**start** and ~~Stop~~**stop** patterns of PDF417 or the ~~Row Address Indicators~~**row address indicators** of MicroPDF417).

The entire symbol shall be scanned multiple times until three conditions are met:

- ~~1a)~~ the maximum number of codewords that ~~could have been~~**can be** decoded is at least ten times the number of codewords in the symbol,
- ~~2b)~~ the highest and lowest decodable rows (which ~~may~~**is** not necessarily ~~be~~ the first and last rows) of the symbol have each been scanned at least three times, and
- ~~3c)~~ at least ~~{0,9;n}~~ of the codewords (data or error correction) have been successfully decoded two or more times, where *n* is the number of non-error-correction data codewords in the symbol.

EXAMPLE Taking a PDF417 symbol with 6 rows and 16 columns and error correction level 4, the total number of codewords is 96, of which 64 are data and 32 error correction. To fulfil condition ~~1,a)~~, the maximum number of codewords that ~~could have been~~**can be** decoded ~~must be~~**is** at least 960. To fulfil condition ~~3,c)~~, since *n* is 64, at least 58 of the codewords ~~must have been~~**to be** decoded twice or more ($0,9 \times 64 = 57,6$).

If the ratio of the total number of validly decoded codewords to the total number of detected row crossings is less than 10:1, then discard the measurements just obtained, and repeat the measurement process, adjusting the tilt angle of the scan line to reduce the number of row crossings.

Otherwise, to compensate for any residual tilt, subtract the number of detected row crossings from the calculated maximum number of codewords that ~~could have been~~**can be** decoded.

Codeword ~~Yield~~**yield** shall be graded from 4,~~0~~ to 0, rounded down to the nearest 0,1, in between grade levels ~~according to~~**in accordance with** Table 2. For example, for ~~Codeword Yield~~**codeword yield** of 0,70 shall result in a grade of 3,8 and a ~~Codeword Yield~~**codeword yield** of 0,54 shall result in a grade of 1,5.

Table 2 — Grading of ~~Codeword Yield~~codeword yield****

Codeword Yield codeword yield	Grade
---	-------

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≥71 %	4,0
≥ 64 % and <71 %	3,0 -to 3,9
≥57 % and <64 %	2,0 -to 2,9
≥50 % and <57 %	1,0 -to 1,9
≥40 % and <50 %	0,0 -to 0,9
<40 %	0,0

6.2.4 Grade based on unused error correction

Decode the symbol completely and process scans until the number of decoded codewords stabilises. Calculate the ~~unused error correction as~~ $1,0 - ((e + 2t) / E_{cap})$, where ~~e is the number of erasures, t is the number of errors and E_{cap} is the error correction capacity of the symbol (the number of error correction codewords minus the number of error correction codewords reserved for error detection).~~ ~~UEC as~~

$$1,0 - [(e + 2t) / E_{cap}]$$

where

- ~~e~~ is the number of erasures;
- ~~t~~ is the number of errors;
- ~~E_{cap}~~ is the error correction capacity of the symbol (i.e. the number of error correction codewords minus the number of error correction codewords reserved for error detection).

If no error correction has been applied to the symbol, and if the symbol decodes, unused error correction shall be taken as 1. If $(e + 2t)$ is greater than E_{cap} , unused error correction shall be taken as 0. In symbols with more than one (e.g. interleaved) error correction block, ~~the~~ unused error correction shall be calculated for each block independently and the lowest value shall be used for grading purposes. ~~Unused~~ ~~The unused~~ error correction shall be graded as shown in Table 3 on a scale from 4,0 to 0 rounded down to the nearest 0,1.

Table 3 — Grading of ~~Unused Error Correction~~ ~~unused error correction~~

Unused Error Correction error correction	Grade
≥0,62	4,0
≥0,50 and <0,62	3,0 -to 3,9
≥0,37 and <0,50	2,0 -to 2,9
≥0,25 and <0,37	1,0 -to 1,9
≥0,20 and <0,25	0,0 -to 0,9
<0,20	0,0

NOTE — ~~Unused error correction is sometimes abbreviated as “UEC”.~~

6.2.5 Grade based on codeword print quality

The approach detailed in this subclause provides additional diagnostic information and enables allowance to be made for the effect of error correction in masking less than perfect attributes of the symbol that influence symbol quality, by applying an overlay technique as described in Annex E. It enables the ~~Decodability~~,

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~~Defects~~~~decodability, defects~~ and ~~Modulation~~~~modulation~~ parameters of scan reflectance profiles covering the entire data region of the symbol to be graded in accordance with ISO/IEC 15416.

This approach uses the following procedure for the assessment of each of the three parameters. In symbols with more than one (e.g. interleaved) error correction block, it shall be applied to each block independently and the lowest value shall be used for grading purposes.

The entire symbol shall be scanned until $0,9 \cdot n$ codewords (where n ~~has~~ ~~the same meaning as~~ ~~number of non-error-correction data codewords in 6.2.3~~ ~~the symbol~~) have been decoded ten times or until it is certain that each codeword has been scanned at least once without inter-row interference. In each scan, the ~~Decodability, Defects~~~~decodability, defects~~ and ~~Modulation~~~~modulation~~ parameters shall be measured in each symbol character in accordance with ISO/IEC 15416. The calculation of all three parameters shall be based on the value of ~~Symbol Contrast~~~~symbol contrast~~ obtained from $R_{\max}$ and $R_{\min}$ in that scan line. The interim codeword grade of each parameter (~~Modulation, Defects~~~~modulation, defects~~ and ~~Decodability~~~~decodability~~) for each codeword is the highest codeword grade for that parameter obtained on any scan for that codeword.

Where the rows include overhead characters (other than the ~~Start~~~~start~~ and ~~Stop~~~~stop~~, or equivalent patterns), for example ~~Row Indicators~~~~row indicators~~ in PDF417 symbols, that are not included in the error correction calculation, these overhead characters shall be assessed first for each row together with the corresponding characters from the rows immediately above and below the row being considered. The highest interim codeword grade for any of these six (or four, in the case of the top or bottom row) characters shall be the overhead grade used to moderate the interim codeword grades for the codewords in the row. If a data codeword's interim codeword grade is higher than the grade obtained by the overhead characters, the data codeword's interim codeword grade shall be reduced to the overhead grade. The interim parameter grades so obtained shall then be modified to allow for the influence of error correction, as described ~~below in the next paragraph, and expounded upon in Annex E and in Table 4.~~

For each parameter, the cumulative number of symbol characters achieving each grade from 4,0 to 0 in steps of 0,1 or a higher grade, and those not decoded, shall be counted, and the counts shall be compared with the error correction capacity of the symbol ~~as follows:~~

~~For each of these 41 grade levels, assuming that all symbol characters not achieving that grade or a higher grade are erasures, derive a notional grade shall be derived for Unused Error Correction~~~~unused error correction~~, on a scale of 4,0 to 0 in steps of 0,1, as described in 6.2.4, based on the percentage thresholds shown in Table 3. The codeword interim grade for each grade level shall be the lower of the grade level and the notional unused error correction grade. The final codeword parameter grade for the symbol shall be the highest codeword interim grade for all grade levels.

NOTE 1 This notional grade is not related to, and does not affect, the unused error correction grade for the symbol as calculated according to 6.2.4, but ~~it~~ is a means of compensating for the extent to which error correction can mask imperfections in a symbol. If one symbol has higher error correction capacity than another symbol, then the former symbol can tolerate a greater number of codewords with poor values for the parameter in question than the latter ~~symbol~~. See Annex E for a ~~fuller~~~~more detailed~~ description of the approach.

Table 4 shows an example of grading one parameter in a symbol containing 100 symbol characters (codewords) with an error correction capacity of 32 codewords. The 100 codewords consist of 68 data codewords, 3 error correction codewords reserved for error detection, and 29 error correction codewords to be used for correcting erasures or errors, giving an erasure correction capacity of 29. The final codeword parameter grade would be 1,2 for the parameter concerned (the highest value in the right-hand column).

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NOTE 2 A similar calculation is performed for ~~each of the parameters Modulation, Defects and Decodability~~parameter modulation, defects and decodability.

Table 4 — Example of codeword print quality parameter grading in symbols with cross-row scanning ability, ~~applying with the~~ overlay procedure in Annex E

MOD/Defects/Decodability effect/decodability grade level (a)	No. Number of codewords at grade level a	no. Cumulative number of codewords at grade level a or higher (b)	Remaining codewords (treated as erasures) (100 – b) (c)	unused error correction UE C capacity (29 – c)	Notional UEC (%)	Notional UEC grade (d)	Codeword interim grade (lower lower of a or d) (e)
<u>a</u>		<u>b</u>	<u>c</u>		<u>%</u>	<u>d</u>	<u>e</u>
4	30	30	70	(exceeded)	<0	0	0
3,9	6	36	64	(exceeded)	<0	0	0
3,8	1	37	63	(exceeded)	<0	0	0
3,7	1	38	62	(exceeded)	<0	0	0
3,6	0	38	62	(exceeded)	<0	0	0
3,5	1	39	61	(exceeded)	<0	0	0
3,4	0	39	61	(exceeded)	<0	0	0
3,3	0	39	61	(exceeded)	<0	0	0
3,2	1	40	60	(exceeded)	<0	0	0
3,1	2	42	58	(exceeded)	<0	0	0
3,0	3	45	55	(exceeded)	<0	0	0
2,9	4	49	51	(exceeded)	<0	0	0
2,8	4	53	47	(exceeded)	<0	0	0
2,7	3	56	44	(exceeded)	<0	0	0
2,6	2	58	42	(exceeded)	<0	0	0
2,5	1	59	41	(exceeded)	<0	0	0
2,4	1	60	40	(exceeded)	<0	0	0
2,3	0	60	40	(exceeded)	<0	0	0
2,2	0	60	40	(exceeded)	<0	0	0
2,1	0	60	40	(exceeded)	<0	0	0
2,0	1	61	39	(exceeded)	<0	0	0
1,9	2	63	37	(exceeded)	<0	0	0
1,8	3	66	34	(exceeded)	<0	0	0
1,7	3	69	31	(exceeded)	<0	0	0
1,6	2	71	29	0	0	0	0
1,5	1	72	28	1	3,45	0	0
1,4	0	72	28	1	3,45	0	0
1,3	5	77	23	6	20,7	0	0
1,2	4	81	19	10	34,5	1,7	1,2

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1,1	5	86	14	15	51,7	3,1	1,1	
1,0	1	87	13	16	55,2	3,4	1,0	
0,9	0	87	13	16	55,2	3,4	0,9	
0,8	0	87	13	16	55,2	3,4	0,8	
0,7	0	87	13	16	55,2	3,4	0,7	
0,6	0	87	13	16	55,2	3,4	0,6	
0,5	0	87	13	16	55,2	3,4	0,5	
0,4	0	87	13	16	55,2	3,4	0,4	
0,3	0	87	13	16	55,2	3,4	0,3	
0,2	0	87	13	16	55,2	3,4	0,2	
0,1	0	87	13	16	55,2	3,4	0,1	
0	0	87	13	0,00	55,2	3,4	0	
Not not decoded	13	100	0	29	100	4	0	
				Parameter grade (Highest highest value of e) _i :			1,2	

~~6.2.5~~6.2.6 Overall symbol grade

The overall symbol grade shall be the lowest of the grade based on analysis of the scan reflectance profile in accordance with 6.2.2, and the grades based on ~~Codeword Yield, Unused Error Correction~~codeword yield, unused error correction and each codeword parameter grade (~~Decodability, Modulation~~decodability, modulation and ~~Defects~~defects) in accordance with 6.2.3, 6.2.4 and 6.2.5. ~~Figure 3 summarizes this process.~~
~~The flowchart in Figure 3 is an informative summary of the process.~~

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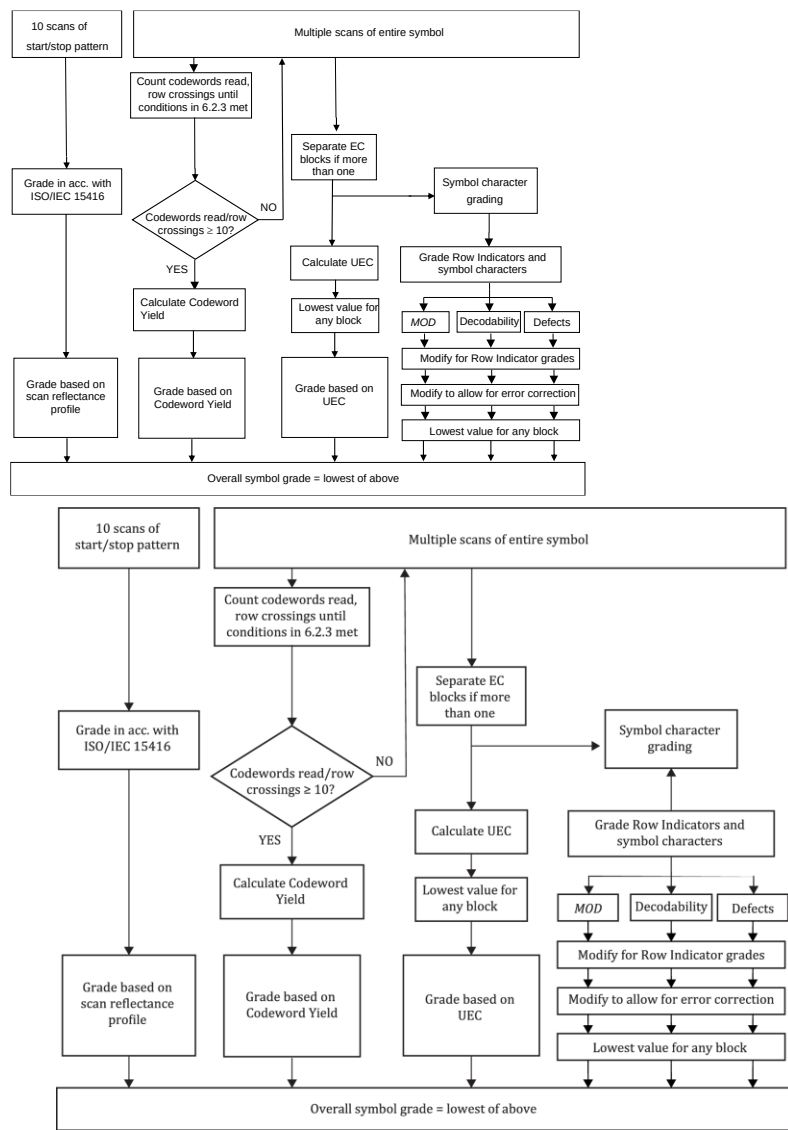


Figure 3 — Grading process for multi-row symbols with cross-row scanning ability

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6.3 Symbolologies requiring row-by-row scanning

The distinguishing feature of these symbolologies is that they require a scan line to traverse a complete row from start to stop pattern (or in the reverse direction) without crossing into an adjacent row and that they require all rows to be scanned.

Each row shall be evaluated in accordance with ISO/IEC 15416 as though it were a separate symbol. Scan lines shall pass through the inspection band of the central 80 % of the height of each row, as specified in ISO/IEC 15416, in order to minimise the effects of cross-talk from adjacent rows. The number of scans per row should be the lower of ten, or the inspection band height divided by the aperture diameter. The overall symbol grade shall be the lowest overall grade obtained for any row.

7 Measurement methodology for two-dimensional matrix symbols

7.1 Overview of methodology

The measurement methodology defined in this clause is based on reflectivity and dimensional measurements and is intended to correlate with conditions encountered in two-dimensional matrix scanning systems.

The method starts by obtaining the raw image, as described in 5.6, which is a high-resolution grey-scale image of the symbol captured under controlled illumination and viewing conditions. The raw image is then converted into a reference grey-scale image, by convolving the raw image with a specified aperture. A binarised image is produced from the reference grey-scale image by applying a specially computed ~~Threshold~~threshold (see Annex A), and this binarised image is then analysed by the reference decode algorithm of the applicable symbology to determine a grid of module centres. The grades for the parameters: ~~Decode~~, ~~Axial Nonuniformity~~, ~~Grid Nonuniformity~~,~~decode~~, ~~axial nonuniformity~~, ~~grid nonuniformity~~ and ~~Unused Error Correction~~unused error correction, are determined from the results of the reference decode algorithm. The grades for the parameter ~~Symbol Contrast~~symbol contrast is measured from the reference grey-scale image within the symbol region including its quiet zones that is determined by the reference decode algorithm. The reference grey-scale image is examined at the module centre locations determined by the reference decode algorithm to obtain values used for grading the parameters: ~~Modulation~~modulation and ~~Fixed Pattern Damage~~fixed pattern damage. These steps are ~~more explicitly defined~~described in greater detail in ~~the following sub-clauses of this document~~7.2 to 7.6.

NOTE The previous ~~revision~~edition of this document recognised possible extreme reflectance values in the neighbourhood of the symbol, which ~~might~~can interfere with reading; however, only their presence was indicated in the report of the overall symbol grade by the use of an asterisk (*) character. This ~~revision~~edition of this document removes the process of recognizing and indicating the presence of extreme reflectance and recommends the use of ISO/IEC 29158 in applications in which there is concern regarding the effect of extreme reflectance on reading performance.

In addition, print growth or loss may be measured and graded according to 7.5.10.

The symbol grade is the lowest grade achieved for these parameters and any others specified for a given symbology or application.

7.2 Test images

7.2.1 Raw image

The raw image consists of the actual reflectance values for each pixel of the light-sensitive array, from which are derived the reference grey-scale image and the binarised image.

7.2.2 Reference grey-scale image

The reference grey-scale image is obtained from the raw image by processing the individual pixel reflectance values through a synthetic aperture as defined in 5.6.2. It is used for the calculation of a ~~Threshold~~threshold and then to generate the binarized image.

7.2.3 Binarised image

The binarised image is obtained from the reference grey-scale image by applying a ~~Threshold~~threshold determined ~~as defined~~ in Annex A. It is analysed by the reference decode algorithm of the applicable symbology to determine a grid of module centres and for the assessment of the parameters ~~Decode, Axial Nonuniformity, Grid Nonuniformity, and Unused Error Correction~~decode, axial nonuniformity, grid nonuniformity and unused error correction.

NOTE The ~~Threshold~~threshold is an optimized value based on a histogram of reflectance levels, not the mid-point between R_{max} and R_{min} as previously defined in ~~an earlier revision~~a past edition of this document.

7.3 Reference reflectivity measurements

7.3.1 Application specification defines the aperture size

An application specification should specify an aperture size as defined in 5.4. An aperture size in the range of 50 % to 80 % of the smallest X dimension to be encountered in an application is recommended (but in all cases is defined by the application specification). In an application where symbols of differing X dimensions will be encountered, the application specification should either:

~~(1a)~~ensure that all measurements are made with the aperture appropriate to the smallest X dimension to be encountered, ~~or~~

~~or~~

~~(2b)~~ specify more than one acceptance criteria with a different aperture size in each acceptance criteria.

An application specification has full discretion to define any number of acceptance criteria. This is normally the symbol grade along with other conditions defined in 5.4. When more than one criterion (~~e.g.~~ with different apertures and ~~a~~ minimum allowed grade ~~for example~~) is offered in an application specification, the specification shall define the acceptance logic. One option is to accept the symbol when one criterion passes. The other option is that both (or more) specified criteria shall pass. However, it is not advisable for an application specification to allow more than one lighting angle and wavelength because those physical attributes of a verifier are not mutable.

NOTE 1 Since the synthetic aperture can be modified in software in practical ~~Verifier~~verifier implementations, this allows a higher flexibility of synthetic aperture definitions within an application specification.

Examples —

~~Aceptance~~EXAMPLE 1The acceptance criteria using an aperture size appropriate to the smallest X dimension to be encountered is:

1,5/08/660/45

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In this case, a grade of 1,5 or higher must be obtained with an aperture of 08.

~~Acceptance~~ **EXAMPLE 2** The acceptance criteria using a variable aperture dependent upon the measured nominal module size is:

1,5/80 %/660/45Q

In this case, an aperture of 80 % of the measured Z dimension is to be used.

~~Acceptance~~ **EXAMPLE 3** The acceptance criteria using two aperture sizes with different minimum acceptable grades is:

1,5/10/660/45 OR 0,5/20/660/45

In this case, a symbol that obtains a grade of either ~~(1)~~

a) a grade of 1,5 or higher measured with an aperture of 10, or ~~(2)~~

b) a grade of 0,5 or higher measured with an aperture of 20 is acceptable.

NOTE 2 In applications in which a single aperture size or lighting does not adequately predict the scanning performance, the use of ISO/IEC 29158 may be ~~preferred~~ chosen over this document alone.

7.4 Grading procedure

The grading procedure is as follows.

- Centre the symbol in the field of view or region of interest.
- Capture the raw image (see 7.2.1).
- Apply the aperture defined in 5.6.2 to the raw image to create a reference grey-scale image (see 7.2.2).
- A circular area in the reference grey-scale image with a diameter 20 times the aperture diameter, centred in the region of interest, shall be used to find the initial ~~Threshold~~ threshold in accordance with Annex A.
- Create a binarised image (see 7.2.3), find the symbol and perform an initial decode.
- Once the symbol has been decoded, compute a new ~~Threshold~~ threshold in accordance with Annex A within the reference grey scale of the image. The area taken into account is only the symbol including its quiet zones. Also compute a revised, or final, $R_{\max}$ and $R_{\min}$, according to 7.5.3 within the inspection area of the reference grey-scale image.
- Create a new binarised image using this new ~~Threshold~~ threshold. Perform a final decode using the applicable reference decode algorithm to determine the definitive grid of module centres for the symbol.
- Calculate all ~~of the~~ graded parameters of the symbol, based on ~~the information determined~~ in the foregoing steps.

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7.5 Image assessment parameters and grading

7.5.1 Use of reference decode algorithm

The symbology reference decode algorithm found in the symbology specification is to be used in the verification process. The reference decode performs the following tasks needed for subsequent measurement of the symbol quality parameters.

- It locates and defines the area covered by the test symbol in the image.
- It determines reference points from the fixed patterns of the symbol, typically the extreme corners, to be used in constructing an ideal grid for measuring GNU.
- It adaptively creates a grid mapping of the data module nominal centres so as to sample them.
- It determines the nominal grid centre spacings in each axis of the symbol (the symbol X dimension).
- It performs error correction, detecting if symbol damage has consumed any of the error capacity.
- It attempts to decode the symbol.

These functions each facilitate the determination of one or more of the parameters described in ~~the following subclauses 7.5.2 to 7.5.11.~~

If this document is applied to a symbology for which there is no standard reference decode algorithm, then this should be noted in the verification report, along with the overall grade.

7.5.2 Decode

The ~~Decoded~~decode parameter tests, on a ~~Pass/Fail~~pass or fail basis, whether the symbol has all its features sufficiently correct to be readable by the reference decode algorithm. The symbology reference decode algorithm shall be used to decode the symbol by computing the module centre positions on the grid determined by processing the binarised image.

If the image cannot be decoded using the symbology reference decode algorithm (even if a proprietary algorithm can be used to decode it), then it shall receive the failing grade 0. In the case of successful decode, codewords are checked to confirm that the codewords are valid according to the symbology specifications (e.g. invalid order, pad characters or use of undefined values are not valid). If any invalid encodation is detected, the decode shall also receive grade 0. Otherwise, it shall receive the grade 4.

7.5.3 Computing $R_{\max}$ and $R_{\min}$

~~Within clause 7 of this document, $R_{\max}$ and $R_{\min}$ shall be defined as follows. For the purposes of Clause 7, $R_{\min}$ shall be computed as the average of the darkest 1 % of the pixels and $R_{\max}$ shall be computed as the average of the brightest 1 % of the pixels in the inspection area.~~

NOTE These ~~definitions~~descriptions of $R_{\max}$ and $R_{\min}$ differ from those in the previous ~~version~~edition of this document and from the ~~definitions~~descriptions in ~~clause~~Clause 6 of this document and are; ~~they apply only to be used on 2D matrix symbols according to clause~~Clause 7 of this document only.

7.5.4 Symbol ~~Contrast~~contrast

The ~~Symbol Contrast~~symbol contrast parameter measures the two reflective states in the symbol, namely light and dark, and the grade indicates the degree to which they are distinct within the symbol.

Symbol contrast, ~~often abbreviated as (SC),~~ is the difference between $R_{\max}$ and $R_{\min}$ values ~~which are determined as described in chapter 7.5.3, defined by:~~

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$\Delta R_{SC} = R_{max} - R_{min}$

Symbol contrast shall be graded on an interpolated scale from 4,0 to 0 and rounded down to the nearest 0,1 as shown in Table 5.

Table 5 — Symbol ~~Contrast~~**contrast** grading

Symbol Contrast contrast	Grade
≥70 %	4,0
≥55 % and <70 %	3,0 to 3,9
≥40 % and <55 %	2,0 to 2,9
≥20 % and <40 %	1,0 to 1,9
≥15 % and <20 %	0 to 0,9
<15 %	0,0

~~Note~~ — Symbol Contrast is sometimes abbreviated as “SC”.

7.5.5 Modulation and related measurements

7.5.5.1 Modulation

Modulation is a measure of the uniformity of reflectance of the dark and light modules respectively. Factors such as print growth ~~(or loss)~~, misplacement of a module relative to the grid intersection, the optical characteristics of the substrate and uneven printing ~~may can~~ reduce the absolute value of the difference between the reflectance of a module and the ~~Threshold~~**threshold**. A low ~~Modulation may~~**modulation can** increase the probability of a module being incorrectly identified as dark or light.

For symbologies that utilize error correction, some incorrectly identified modules may be tolerated within the error correction capacity of the symbol, so the grade assigned to ~~Modulation~~**modulation** takes this into account. Furthermore, modules that have reflectivity close to the ~~Threshold~~**threshold** and therefore may have higher probability to be incorrectly identified as dark or light, are also taken into account. Therefore, grading of ~~Modulation~~**modulation** is closely related to the grading of ~~Unused Error Correction~~**unused error correction**, as explained below and in Annex E. While ~~Modulation~~**modulation** is graded only considering the data modules within the ~~Symbol~~**symbol**, the values of MOD for modules within the fixed pattern are used to grade fixed pattern grading.

The value of MOD for each module is calculated from the reference grey-scale image value, *R*, at each module centre (determined by applying the symbology reference decode algorithm to the binarised image), *R*_{max}, *R*_{min} and the ~~Threshold~~**threshold** as follows:

— ~~for modules whose correct state is light:~~

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- For modules whose correct state is light:

$$M_{MOD} = (R - T) / (R_{max} - T) \text{ for } R \geq T \text{ and } R < R_{max}$$

$$M_{MOD} = 100 \text{ for } R \geq R_{max}$$

$$M_{MOD} = 0 \text{ for } R < T$$

and for modules whose correct state is dark:

- for modules whose correct state is dark:

$$M_{MOD} = (T - R) / (T - R_{min}) \text{ for } R < T \text{ and } R \geq R_{min}$$

$$M_{MOD} = 100 \text{ for } R < R_{min}$$

$$M_{MOD} = 0 \text{ for } R \geq T$$

- Where

M_{MOD} is modulation value for the module

- R is the reflectance of the module, given by the reference grey scale image at the module centre

- T is the Threshold computed by Annex A over all pixels within the symbol area

- R_{max} is the average maximum reflectance determined in accordance with 7.5.3

- R_{min} is the average minimum reflectance determined in accordance with 7.5.3

where

- M_{MOD} is the modulation value for the module;

- R is the reflectance of the module, given by the reference grey scale image at the module centre;

- T is the threshold computed by Annex A over all the pixels within the symbol area;

- R_{max} is the average maximum reflectance determined in accordance with 7.5.3;

- R_{min} is the average minimum reflectance determined in accordance with 7.5.3.

Assign the grade level for each module according to in accordance with Table 6. For each codeword, select the minimum modulation grade of all modules in the codeword. Since M_{MOD} will be zero for all modules which are detected as a module error (that is, i.e. R higher than T for a dark module or R is lower than T for a light module), the Module Grade shall also be 0 for those modules.

Table 6 — Module grading for ~~Modulation~~modulation

MOD	Module Grade
$\geq 0,50$	4,0
$\geq 0,40$ and $< 0,50$	3,0 to 3,9
$\geq 0,30$ and $< 0,40$	2,0 to 2,9
$\geq 0,20$ and $< 0,30$	1,0 to 1,9
$\geq 0,175$ and $< 0,20$	0,0 to 0,9
$< 0,175$	0,0

Since grades are assigned on a scale from 4,0 to 0 in steps of 0,1, there are 41 possible grade levels. The cumulative number of codewords achieving each grade shall be counted and compared with the error correction capacity of the symbol as follows:

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For each grade level, assuming that all codewords not achieving that grade or a higher grade are errors, derive a notional ~~Unused Error Correction~~unused error correction grade as described in 7.5.9. Take the lower of the grade level and the notional UEC grade.

NOTE 1 This notional grade does not affect the UEC grade for the symbol as calculated according to 7.5.9, but is a means of compensating for the extent to which error correction can mask imperfections in a symbol. If one symbol has higher error correction capacity than another symbol, then the former symbol can tolerate a greater number of codewords with low modulation than the latter. See Annex E for a fuller description of the approach.

Then the ~~Modulation~~modulation grade for the symbol shall be the highest of the resulting values for all grade levels. When the symbol consists of more than one (e.g. interleaved) error correction block, each block shall be assessed independently and the lowest grade for any block shall be taken as the ~~Modulation~~modulation grade of the symbol.

Tables 7 and 8 show examples of grading ~~Modulation~~modulation in a ~~Data Matrix~~data matrix symbol containing 18 codewords, 10 of which are error correction codewords with a capacity to correct up to 5 errors in a single error correction block. The ~~Modulation~~modulation grade of the symbol in the example of Table 7 is 2,2 (the highest value in the right-hand column) because that is the notional UEC grade for all levels of M_{MOD} codeword grade greater than 2,2. The ~~Modulation~~modulation grade in the example of Table 8 is 3,4 because the notional UEC grade is greater than 3,4 for all MOD levels below 3,4.

NOTE 2 The previous ~~revision~~edition of this document defined ~~Reflectance Margin~~reflectance margin and ~~Modulation~~modulation. This ~~version~~edition of this document merges these into one parameter called ~~Modulation~~modulation.

Table 7 — Example of ~~Modulation~~modulation grading in a ~~two-dimensional~~2D matrix symbol, applying with the overlay procedure in Annex E

MOD codeword grade level (a)	No. Number of codewords at level a	Cumulative no. number of codewords at level a or higher (b)	Remaining codewords (treated as errors) (18 - b) (c)	Notional unused error correction UEC capacity (10 - c)	Notional UEC	Notional UEC grade (d)	Lower of Codeword interim grade (lower than a or d) (e)
a		b	c			d	e
4,0	10	10	8	Exceeds exceeds	<0	0	0
3,9	5	15	3	2	0,40	2,2	2,2
3,8	0	15	3	2	0,40	2,2	2,2
3,7	0	15	3	2	0,40	2,2	2,2
3,6	0	15	3	2	0,40	2,2	2,2
3,5	0	15	3	2	0,40	2,2	2,2
...	...	...	...	...	...	...	...
(21 Rows: rows: 3,4 to 1,4)	(0 in each row)	(15 in each row)	(3 in each row)	(2 in each row) each row)	(0,40 in each row)	(2,2 in each row)	(2,2 in each row)
1,3	0	15	3	2	0,40	2,2	1,3
1,2	0	15	3	2	0,40	2,2	1,2

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1,1	0	15	3	2	0,40	2,2	1,1
1,0	0	15	3	2	0,40	2,2	1,0
0,9	1	16	2	3	0,60	3,8	0,9
0,8	0	16	2	3	0,60	3,8	0,8
0,7	1	17	1	4	0,80	4,0	0,7
0,6	0	17	1	4	0,80	4,0	0,6
0,5	0	17	1	4	0,80	4,0	0,5
0,4	0	17	1	4	0,80	4,0	0,4
0,3	0	17 7	1	4	0,80	4,0	0,3
0,2	0	17	1	4	0,80	4,0	0,2
0,1	0	17	1	4	0,80	4,0	0,1
0	1	18	0	5	1,00	4,0	0
					Modulation grade (Highest value of e):		2,2

Table 8 — Example of ~~Modulation~~modulation grading in a ~~two-dimensional~~2D matrix symbol, applying with the overlay procedure in Annex E

MOD codeword grade level (a)	No. Number of codewords at level a	Cumulative no. number of codewords at level a or higher (b)	Remaining codewords (treated as errors) (18 - b) (c)	Notional unused error correction capacity (5 - c)	Notional UEC	Notional UEC grade (d)	Lower of Codeword interim grade (lower than a or d) (e)
a		b	c			d	e
4,0	10	10	8	Exceeded	<0	0	0
3,9	5	15	3	2	0,60	2,2	2,2
3,8	0	15	3	2	0,60	2,2	2,2
3,7	0	15	3	2	0,60	2,2	2,2
3,6	0	15	3	2	0,60	2,2	2,2
3,5	0	15	3	2	0,60	2,2	2,2
3,4	3	18	0	5	1,00	4,0	3,4
3,3	0	18	0	5	1,00	4,0	3,3
3,2	0	18	0	5	1,00	4,0	3,2
... (28 rows: 3,1 ... 0,4)	... (0 in each row)	... (18 in each row)	... (0 in each row)	... (5 in each row)	... (1,00 in each row)	... (4,0 in each row)	... (3,1 to 0,4)
0,3	0	18	0	5	1,00	4,0	0,3
0,2	0	18	0	5	1,00	4,0	0,2
0,1	0	18	0	5	1,00	4,0	0,1
0	0	18	0	5	1,00	4,0	0

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					Modulation grade (Highest highest value of e):	3,4	
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7.5.5.2 Reporting ~~Contrast Uniformity~~contrast uniformity

Take the lowest value of M_{MOD} of any module in the data region of the symbol, as computed in 7.5.5.1 and report it as ~~Contrast Uniformity~~contrast uniformity (CU). This value is not graded, but only used for conformance testing and for process control, because it is the worst-case modulation found in any data module within the symbol.

NOTE 1 Contrast ~~Uniformity~~uniformity is sometimes abbreviated as “CU”.

NOTE 2 ~~Contrast Uniformity will be non-zero~~; only when unused error correction is 100 % since otherwise at least one value of M_{MOD} will be zero.

7.5.6 Fixed ~~Pattern Damage~~pattern damage

This parameter grades damage to the finder pattern, quiet zone, timing, navigation and other fixed patterns in a symbol that ~~may can~~ reduce the ability of the reference decode algorithm to locate and identify the symbol within the field of view. The particular patterns to be considered, and the amounts of damage corresponding to the various grade thresholds, must be specified independently for each symbology. Fixed ~~Pattern Damage~~pattern damage (FPD) shall be graded using the rules appropriate to each symbology, using the rules specified in the applicable symbology specification or, for the ~~Maxicode~~maxicode symbology specifically, as specified in ISO/IEC 15415:2011-~~Annex~~, Clause A.3.

Fixed ~~Pattern Damage~~pattern damage is evaluated in terms of the number of modules with low modulation and modules that appear as the inverse of the intended colour in the feature (or part of the feature) concerned. The MOD value of each module in the ~~Fixed Pattern~~fixed pattern is calculated as defined in 7.5.5.1, and is assigned a ~~Module Grade~~module grade (in the range of 0 through 4,0 in steps of 0,1) according to Table 6. ~~In this~~This way, the fixed pattern damage is graded on a continuous scale between 4,0 and 0,0 rounded down to the nearest 0,1.

Note — Fixed Pattern Damage is sometimes abbreviated as “FPD”.

7.5.7 Axial ~~Nonuniformity~~

7.5.7 Axial nonuniformity

Two-dimensional matrix symbols include data fields of modules nominally lying in a regular polygonal grid, and any reference decode algorithm must adaptively map the centre positions of those modules to extract the data. Axial ~~Nonuniformity~~nonuniformity (ANU) measures and grades the spacing of the mapping centres, i.e. the sampling points, or intersections of the grid obtained by applying the reference decode algorithm to the binarised image, in the direction of each of the grid’s major axes. Axial ~~Nonuniformity~~nonuniformity tests for uneven scaling of the symbol which would hinder readability of the symbol.

The spacings between adjacent sampling points are independently sorted for each polygonal axis, then the average spacings X_{AVG} , Y_{AVG} , ... along each axis are computed. Axial ~~Nonuniformity~~nonuniformity is a measure of how much the sampling point spacing differs from one axis to another, namely:

$$M_{ANU} = \text{abs}(X_{AVG} - Y_{AVG}) / ((X_{AVG} + Y_{AVG}) / 2)$$

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where $\text{abs}()$ yields the absolute value.

If a symbology has more than two major axes, ~~then Axial Nonuniformity~~the axial nonuniformity is computed for those two average spacings which differ the most.

Note — Axial Nonuniformity is sometimes abbreviated as “ANU”.

Axial NonuniformityAxial nonuniformity shall be graded as shown in Table 9.

Table 9 — Axial ~~Nonuniformity~~nonuniformity grading

Axial Nonuniformity nonun iformity	Grade
$\leq 0,06$	4,0
$\leq 0,08$ and $> 0,06$	3,0 to 3,9
$\leq 0,10$ and $> 0,08$	2,0 to 2,9
$\leq 0,12$ and $> 0,10$	1,0 to 1,9
$\leq 0,14$ and $> 0,12$	0,0 to 0,9
$> 0,14$	0,0

7.5.8 Grid ~~Nonuniformity~~nonuniformity

Grid ~~Nonuniformity~~nonuniformity (GNU) measures and grades the largest vector deviation of the grid intersections, determined by the reference decode algorithm from the binarized image of a given symbol, from their ideal theoretical position.

NOTE 1 — Grid Nonuniformity is sometimes abbreviated as “GNU”.

Using the reference decode algorithm for the symbology, plot the positions of all grid intersections in the data area of the symbol and compare these positions with the ideal grid in a theoretical perfect symbol of the same nominal dimensions. The ideal grid is evenly spaced and based on the minimum number of points that defines the symbols borders or axes. The symbology specification should define these points for the purpose of grading ~~Grid Nonuniformity~~the grid nonuniformity. The greatest distance between the actual and the theoretical position of any intersection, expressed as a fraction of the X dimension of the symbol, shall be taken for grading purposes.

Grid ~~Nonuniformity~~nonuniformity shall be graded as shown in Table 10.

Table 10 — Grid ~~Nonuniformity~~nonuniformity grading

Grid Nonuniformity nonunifo rmity	Grade
$\leq 0,38$	4,0
$\leq 0,50$ and $> 0,38$	3,0 to 3,9
$\leq 0,63$ and $> 0,50$	2,0 to 2,9
$\leq 0,75$ and $> 0,63$	1,0 to 1,9
<u>$> 0,75$</u>	<u>0,0</u>

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> 0,75 NOTE The GNU grade is zero whenever the GNU value exceeds 0,75 with no gradual change between 1,0 and 0.

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NOTE 2 — GNU grade is zero whenever the GNU value exceeds 0,75 with no gradual change between 1,0 and 0.

7.5.9 Unused error correction

The ~~Unused Error Correction~~ unused error correction parameter tests the extent to which regional or spot damage in the symbol has eroded the reading safety margin that error correction provides. Unused error correction is graded based on the amount of error and erasures found by the reference decode algorithm. The amount of ~~Unused Error Correction~~ unused error correction is calculated as

$$M_{\text{UEC}} = 1,0 - ((e + 2t) / E_{\text{cap}}),$$

where

e is the number of erasures;

t is the number of errors;

E_{cap} is the error correction capacity of the symbol (i.e. the number of error correction codewords minus the number of error correction codewords reserved for error detection).

If no error correction has been applied to the symbol, and if the symbol decodes, unused error correction is taken as 1. If $(e + 2t)$ is greater than E_{cap} , unused error correction is taken as 0. In symbols with more than one (e.g. interleaved) error correction block, unused error correction shall be calculated for each block independently and the lowest value shall be used for grading purposes.

NOTE — Unused error correction is sometimes abbreviated as “UEC”.

~~Unused Error Correction~~ shall be graded as shown in Table 11.

Table 11 — Unused ~~Error Correction~~ error correction grading

Unused error correction	Grade
$\geq 0,62$	4,0
$\geq 0,50$ and $< 0,62$	3,0 to 3,9
$\geq 0,37$ and $< 0,50$	2,0 to 2,9
$\geq 0,25$ and $< 0,37$	1,0 to 1,9
$\geq 0,20$ and $< 0,25$	0,0 to 0,9
$< 0,20$	0,0

7.5.10 Print ~~Growth~~ growth

Print ~~Growth~~ growth (PG) tests that the graphical features comprising the symbol have not grown or shrunk from its nominal state so much as to hinder readability with less favourable imaging conditions than the test condition. The print growth parameter, the extent to which dark or light markings appropriately fill their

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module boundaries, is an important indication of process quality which affects reading performance. Print growth may be measured and evaluated independently in more than one axis, to determine, for example, both horizontal and vertical growth.

Print growth shall be graded according to a symbology specification, if defined. If a symbology specification states that ~~Print Growth~~print growth shall be graded, but does not define a specific method, it shall be graded as defined ~~below in this clause~~. Otherwise, it is not a graded parameter but should be reported as an informative measure for the purposes of process control.

Print ~~Growth may~~growth can be measured in the general way described ~~herein this subclause~~ but ~~may it can~~ be more specifically defined in a symbology specification.

Identify the graphical structures particular to the symbology that are most indicative of element growth or shrinkage in each axis of the symbol, which ~~will are~~ generally ~~be~~ fixed structures or isolated elements, such as clock tracks. Based on the symbology specification and its reference decode algorithm, determine for each of these structures, in each axis, its nominal dimension in modules. Measure the actual dimension in modules, of the structure along the grid lines derived by the use of the reference decode algorithm and passing through each structure to be measured in the symbol axis in question. The difference between the actual dimension and the nominal dimension, as a fraction of a module, is the print growth for each fixed structure. Print growth shall then be calculated for each axis as the arithmetic mean of the print growth of all fixed structure measured as described above. Where the result is negative, it represents print loss.

Unless specified differently in a symbology specification, and where a symbology specification states that ~~Print Growth~~print growth shall be graded in accordance with this document and not defined alternately, the ~~Print Growth~~print growth grade shall be calculated ~~according to in accordance with~~ Table 12 in the range from 0,0 to 4,0 rounded down to the nearest 0,1 by using the absolute value of print growth.

Table 12 — ~~Print Growth Grading (default)~~Default print growth grading

Absolute Valuevalue of Print Growthprint growth	Grade
$\leq 0,10$	4,0
$\leq 0,28$ and $> 0,10$	3,0 to 3,9
$\leq 0,34$ and $> 0,28$	2,0 to 2,9
$\leq 0,40$ and $> 0,34$	1,0 to 1,9
$\leq 0,60$ and $> 0,40$	0,1 to 0,9
$> 0,60$	0,0

NOTE 1 — Print growth is sometimes abbreviated as "PG".

NOTE 2 The recommendation that ~~Print Growth~~print growth be a graded parameter is a new feature of this ~~revision~~edition of this document, see 7.5.11.2.

7.5.11 Additional grading parameters

7.5.11.1 Symbology and General

~~Either symbology~~ or application specifications ~~may, or both, can~~ define additional parameters which may be graded and ~~taken into account~~accounted for in the calculation of the overall symbol grade.

~~7.5.11.1~~7.5.11.2 Application specifications

An application specification ~~may can~~ require, among other things, that the Z dimension is within a certain range or that the data encoded within a symbol meets certain defined criteria.

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7.5.11.3 Symbology specifications

Symbology specifications should define fixed pattern damage grading.

NOTE — Fixed pattern damage is sometimes abbreviated as “FPD”.

It is recommended that symbology specifications define ~~Print Growth~~print growth as a graded parameter because print growth ~~may~~is not ~~be~~always detected by ~~any~~ other graded ~~parameter~~parameters especially when ~~the~~ aperture size is small relative to the *X* dimension (see [Clause F.4](#)). If ~~Print Growth~~print growth is to be graded, but the general approach described in this document for grading ~~Print Growth~~print growth does not fit the symbology, then an alternate definition for measuring and grading of ~~Print Growth~~print growth should be defined in the symbology standard.

Symbology standards must also define the reference decode algorithm to be used for the parameter ~~DECODE~~decode and for defining the symbol grid which is used for evaluating many other parameters, as well as the basis for defining the ideal grid used for computing ~~Grid Nonuniformity~~grid nonuniformity.

7.6 Symbol ~~Grade~~grade

The symbol grade shall be the lowest grade of any parameter as measured in accordance with 7.5.2 through 7.5.11, excluding 7.5.5.2.

In order to determine the causes of ~~grades of~~poor quality ~~grades~~, it is necessary to examine the grades for each parameter, especially the one(s) with the lowest grade(s). Annex B offers some guidance on the interpretation and possible root causes for low grades.

Overall symbol grades shall be expressed on a numeric scale ranging in descending order of quality from 4,0 to 0,0 rounded down to the nearest 0,1.

8 Measurement methodologies for composite symbologies

Each component shall be measured and graded separately. The linear component shall be measured and graded in accordance with ISO/IEC 15416. When the two-dimensional composite component uses a multi-row bar code symbology, then the methodology specified in ~~clause~~Clause 6 shall be applied to the two-dimensional composite component; when it uses a two-dimensional matrix symbology, then the methodology specified in ~~clause~~Clause 7 shall be applied to it. Both the overall grade for the linear component so measured and the overall grade for the two-dimensional composite component shall be reported, to assist users who ~~may~~only require to read the linear component as well as those requiring to read the complete symbol.

Annex A
(normative)

Thresholding ~~Algorithm~~algorithm based on ~~Histogram~~histogram

A.1 Algorithm description

Start by creating a histogram of the defined grey-scale values in the defined region and proceed as follows:

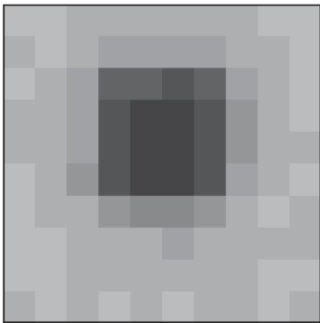
- ~~1.a)~~ Initialize the variable minVariance to a very large number and initialize Tmin and Tmax to zero.
- ~~2.b)~~ For every grey-scale value, "g", starting from the lowest grey-scale value to the highest grey-scale value (0 to 255 for an 8-bit image sensor);
 - ~~a.1)~~ Compute the mean and variance of pixels below g and call it MeanDark and VarianceDark.
 - ~~b.2)~~ Compute the mean and variance of pixels above or equal to g and call it MeanLight and VarianceLight.
 - ~~c.3)~~ Compute Variance = VarianceLight + VarianceDark.
 - ~~d.4)~~ If Variance < minVariance, save Variance in minVariance and save g in Tmin.
 - ~~e.5)~~ If Variance = minVariance save g in Tmax.

NOTE Step ~~e.5~~ is used to break ties. Tmin is the smallest grey-level where the variance is the minimum and Tmax is the largest grey-level where the variance is the same minimum.

- ~~3.c)~~ Optimal threshold $T = (Tmin + Tmax) / 2$.

A.2 Example

For simplicity, an image with only 100 pixels (a 10 ~~x~~ 10 image) ~~will be~~is used. Additionally, for the purpose of the example, the image is composed of 4-bit pixels (16 grey levels). The sample image is shown in Figure A.1, where each pixel is magnified so that individual pixels can be discerned.



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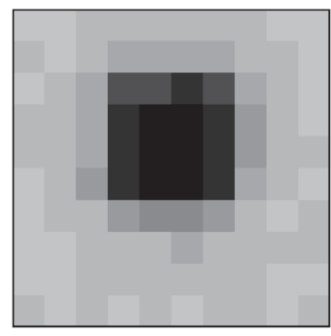


Figure A.1 — Image used in this example.

We begin by counting how many. Firstly, the number of pixels are contained in the image with each of the 16 grey levels are counted. The result of this count is shown in Table A.1, and is plotted as a histogram in Figure A.2.

Table A.1 — Count of grey level occurrences

Grey Levellevel	Number of pixels with grey level
0	0
1	0
2	6
3	7
4	3
5	0
6	0
7	2
8	5
9	10
10	44
11	23
12	0
13	0
14	0

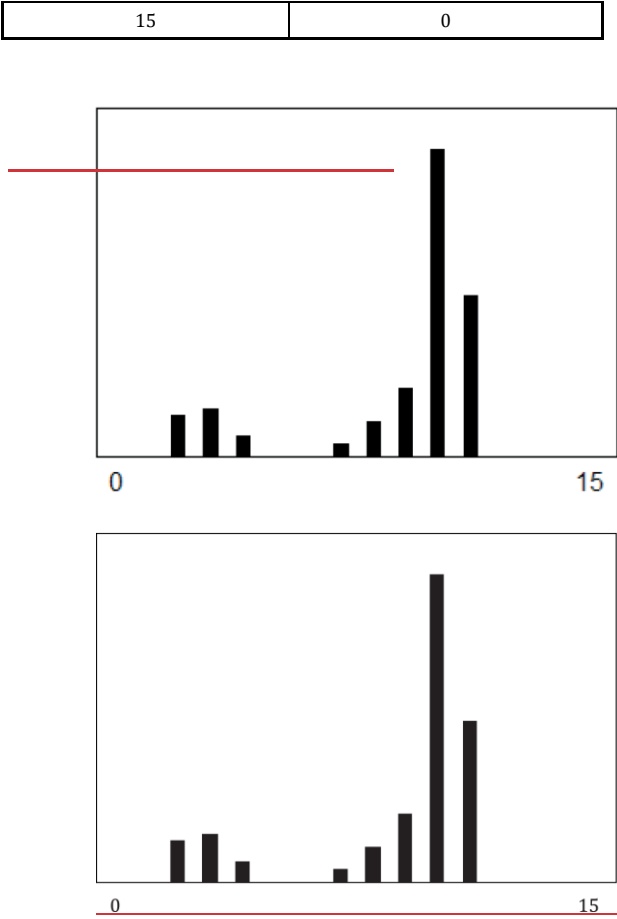


Figure A.2 — Histogram of the data from Table A.1

For each possible threshold, ~~we separate~~ the histogram is separated into two portions – one for the dark elements and one for the light elements. The first possible threshold is between 0 and 1, the next is between 1 and 2, and so on. For each possible threshold, ~~compute~~ the variance of both portions of the histogram. is computed. For example, for the possible threshold between 4 and 5, the dark element histogram contains the grey levels 0, 1, 2, 3 and 4 as shown in Table A.2.

Table A.2 — Dark pixel portion for threshold of 4.5

Grey Level level	Number of pixels with grey level
0	0

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1	0
2	6
3	7
4	3

The variance of this distribution is calculated as follows:

The mean is $\frac{[(2 \times 6) + (3 \times 7) + (4 \times 3)]}{16} = 2.81$, which can be described as the weighted average of Table A.2.

The variance is the average of the square of each element's deviation from the mean:

$$\frac{([2.81 - 2]^2 \times 6) + ([2.81 - 3]^2 \times 7) + ([2.81 - 4]^2 \times 3)}{16} = 0.53$$

Similarly, the variance of the light elements (those whose pixel value is 5 or greater) is: 0.84.

The variances of the dark and light portions of the histogram for each threshold can be computed in a similar manner. The results are shown in Table A.3.

Table A.3 — List of variances for all possible thresholds

Threshold	Variance of Dark Elements	Variance of Light Elements	Sum of Variances
0,5	0,00	7,67	7,67
1,5	0,00	7,67	7,67
2,5	0,00	5,00	5,00
3,5	0,25	2,00	2,25
4,5	0,53	0,84	1,37
5,5	0,53	0,84	1,37
6,5	0,53	0,84	1,37
7,5	2,20	0,65	2,85
8,5	5,52	0,40	5,92
9,5	8,50	0,23	8,73
10,5	8,11	0,00	8,11
11,5	7,67	0,00	7,67
12,5	7,67	0,00	7,67
13,5	7,67	0,00	7,67
14,5	7,67	0,00	7,67

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15,5	7,67	0,00	7,67
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An optimum threshold is chosen such that the sum of variances of both portions of the histogram is minimized. As can be seen from Table A.3, the minimum of the sum of variances is 1.37 which occurs at thresholds: 4.5, 5.5, and 6.5. There is a range of thresholds that all give this minimum variance. In this case take the average of the lowest and highest threshold which gives this minimum which is 5.5 in this example.

Note that the threshold obtained by this averaging ~~will~~does not necessarily have the same minimum sum of variances as it does in this example. If there is a single threshold which gives the minimum sum of variances, then take that threshold. This chosen threshold is considered the “optimum” threshold as determined by the algorithm in Clause A.1, because it results in two separate portions of the overall histogram which are assumed to be most representative of two groups of elements (dark and light).

Note

NOTE The portion of the histogram to the right of the threshold is called the “light lobe”.

When the image is binarized using the calculated threshold, the result is shown in Figure A.3.

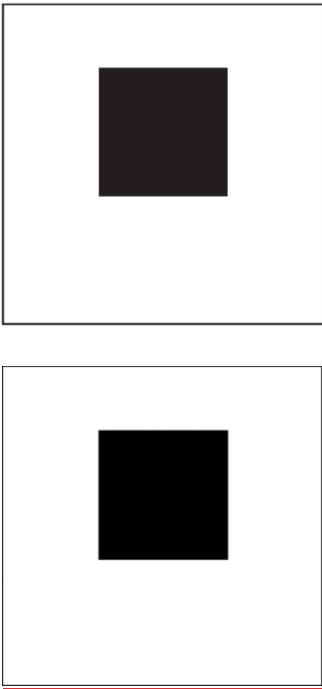


Figure A.3 — Image with threshold applied

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Annex B
(informative)

Interpreting the scan and symbol grades

This Annex describes possible causes of reduced grades, either in a multi-row symbol or a matrix symbol.

~~The table below~~Table B.1 identifies ~~a number of~~ factors that ~~may~~can lead to low or failing grades for the parameters indicated, which ~~may~~can be similar or ~~can~~ differ for the two classes of two-dimensional ~~symbols~~symbols.

Table B.1 — Possible causes of low grades

Parameter	Multi-row symbols	Matrix symbols
Symbol contrast	— low background or light module reflectance, due to:	— low background or light module reflectance, due to:
	— incorrect substrate e.g. blue paper for red light	— incorrect substrate e.g. blue paper for red light
	— glossy laminate/overwrap	— glossy laminate/overwrap
	— inappropriate angle of illumination (direct marked symbols)	— inappropriate angle of illumination (direct marked symbols)
Symbol Contrast	— high dark module reflectance, due to	— high dark module reflectance, due to
	— low absorption of incident light by ink (unsuitable formulation/colour)	— low absorption of incident light by ink (unsuitable formulation/colour)
	— insufficient ink coverage (e.g. non-overlapping ink-jet dots)	— insufficient ink coverage (e.g. non-overlapping ink-jet dots)
	— inappropriate angle of illumination (direct marked symbols)	
	— inappropriate angle of illumination (direct marked symbols)	— inappropriate angle of illumination (direct marked symbols)
Decode	— many factors — see other parameters in this table	— many factors - see other parameters in this table
	— software errors in printing system	— software errors in printing system

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Parameter	Multi-row symbols	Matrix symbols
Symbol contrast	- low background or light module reflectance, due to: - incorrect substrate e.g. blue paper for red light - glossy laminate/overwrap - inappropriate angle of illumination (direct marked symbols)	- low background or light module reflectance, due to: - incorrect substrate e.g. blue paper for red light - glossy laminate/overwrap - inappropriate angle of illumination (direct marked symbols)
Symbol Contrast	- high dark module reflectance, due to - low absorption of incident light by ink (unsuitable formulation/colour) - insufficient ink coverage (e.g. non-overlapping ink-jet dots) - inappropriate angle of illumination (direct marked symbols)	- high dark module reflectance, due to - low absorption of incident light by ink (unsuitable formulation/colour) - insufficient ink coverage (e.g. non-overlapping ink-jet dots)
	inappropriate angle of illumination (direct marked symbols)	inappropriate angle of illumination (direct marked symbols)
Unused Error Correction or correction	- physical damage (scuffing, tearing, obliteration) - bit errors due to defects - excessive print growth in one or two axes - local deformation - misplaced modules	- physical damage (scuffing, tearing, obliteration) - bit errors due to defects - excessive print growth in one or two axes - local deformation - misplaced modules
Minimum Reflectance (reflectance, R_{min})	Reflectance of all bars $>0,5R_{max}$ = see symbol	=
Minimum Edge Contrast edge contrast	- excessive print growth/loss - too large measuring aperture - irregular substrate reflectance - low ink coverage - showthrough	=

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Parameter	Multi-row symbols	Matrix symbols
Symbol contrast	— low background or light module reflectance, due to: — incorrect substrate e.g. blue paper for red light — glossy laminate/overwrap — inappropriate angle of illumination (direct marked symbols)	— low background or light module reflectance, due to: — incorrect substrate e.g. blue paper for red light — glossy laminate/overwrap — inappropriate angle of illumination (direct marked symbols)
Symbol Contrast	— high dark module reflectance, due to — low absorption of incident light by ink (unsuitable formulation/colour) — insufficient ink coverage (e.g. non-overlapping ink-jet dots) — inappropriate angle of illumination (direct marked symbols)	— high dark module reflectance, due to — low absorption of incident light by ink (unsuitable formulation/colour) — insufficient ink coverage (e.g. non-overlapping ink-jet dots)
	— inappropriate angle of illumination (direct marked symbols)	— inappropriate angle of illumination (direct marked symbols)
Modulation	— print growth/loss — too large measuring aperture — irregular substrate reflectance — variation in ink coverage — showthrough	— print growth or loss — too large measuring aperture — misplaced modules — defects (spots or voids) — irregular substrate reflectance — variation in ink coverage — showthrough
Defects	— spots of ink or other dark marks on background — voids in printed areas — faulty print head elements — too small measuring aperture	=

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Parameter	Multi-row symbols	Matrix symbols
Symbol contrast	— low background or light module reflectance, due to: — incorrect substrate e.g. blue paper for red light — glossy laminate/overwrap — inappropriate angle of illumination (direct marked symbols)	— low background or light module reflectance, due to: — incorrect substrate e.g. blue paper for red light — glossy laminate/overwrap — inappropriate angle of illumination (direct marked symbols)
Symbol Contrast	— high dark module reflectance, due to — low absorption of incident light by ink (unsuitable formulation/colour) — insufficient ink coverage (e.g. non-overlapping ink-jet dots) — inappropriate angle of illumination (direct marked symbols)	— high dark module reflectance, due to — low absorption of incident light by ink (unsuitable formulation/colour) — insufficient ink coverage (e.g. non-overlapping ink-jet dots)
	— inappropriate angle of illumination (direct marked symbols)	— inappropriate angle of illumination (direct marked symbols)
Decodability	— local distortion — pixel errors in printing — slippage during printing — blocked inkjet nozzle — faulty thermal element	=
Codeword Yield	— excessive tilt of scan line — Y axis print growth — thermal "drag"	=
Fixed Pattern Damage pattern damage	=	— blocked printer nozzle — faulty thermal element — physical damage (tearing, scuffing, obliteration)

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Parameter	Multi-row symbols	Matrix symbols
Symbol contrast	— low background or light module reflectance, due to: — incorrect substrate e.g. blue paper for red light — glossy laminate/overwrap — inappropriate angle of illumination (direct marked symbols)	— low background or light module reflectance, due to: — incorrect substrate e.g. blue paper for red light — glossy laminate/overwrap — inappropriate angle of illumination (direct marked symbols)
Symbol Contrast	— high dark module reflectance, due to — low absorption of incident light by ink (unsuitable formulation/colour) — insufficient ink coverage (e.g. non-overlapping ink-jet dots) — inappropriate angle of illumination (direct marked symbols)	— high dark module reflectance, due to — low absorption of incident light by ink (unsuitable formulation/colour) — insufficient ink coverage (e.g. non-overlapping ink-jet dots)
	— inappropriate angle of illumination (direct marked symbols)	— inappropriate angle of illumination (direct marked symbols)
Axial Nonuniformity nonuniformity	=	— mismatch of transport speed in printing with symbol dimensions — printing software errors — verifier axis not perpendicular to symbol plane
Grid Nonuniformity nonuniformity	=	— transport errors in printing (acceleration/deceleration, vibration, slippage) — variation in printhead to substrate distance — verifier axis not perpendicular to symbol plane — pixel errors in printing

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Parameter	Multi-row symbols	Matrix symbols
Symbol contrast	— low background or light module reflectance, due to: — incorrect substrate e.g. blue paper for red light — glossy laminate/overwrap — inappropriate angle of illumination (direct marked symbols)	— low background or light module reflectance, due to: — incorrect substrate e.g. blue paper for red light — glossy laminate/overwrap — inappropriate angle of illumination (direct marked symbols)
Symbol Contrast	— high dark module reflectance, due to — low absorption of incident light by ink (unsuitable formulation/colour) — insufficient ink coverage (e.g. non-overlapping ink-jet dots) — <u>inappropriate angle of illumination (direct marked symbols)</u>	— high dark module reflectance, due to — low absorption of incident light by ink (unsuitable formulation/colour) — insufficient ink coverage (e.g. non-overlapping ink-jet dots)
	— inappropriate angle of illumination (direct marked symbols)	— inappropriate angle of illumination (direct marked symbols)
Print Growth/Loss growth or loss	— print process-dependent factors — absorbency of substrate — dot size (ink-jet, dot peening, etc.) — incorrect thermal print head temperature	— print process-dependent factors — absorbency of substrate — dot size (ink-jet, dot peening, etc.) — incorrect thermal print head temperature

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Annex C (informative)

Guidance on ~~selection of~~ selecting grading parameters in application specifications

C.1 Selection of measurement wavelength

C.1.1 General considerations

The wavelength of light used for verification is very important and should be specified by ~~an~~the appropriate application ~~as appropriate~~. This Annex provides guidance to applications with respect to typical sources of illumination and so that verification of symbols ~~may~~can correlate to scanning performance in actual applications. The application standard may select any appropriate source of illumination, whether or not it is listed ~~below in this Annex~~. In the absence of an application specification giving illumination (and other) parameters, verification results are not reproducible.

Clauses 6 and 7 ~~of this document~~ require measurements to be made using light of the same characteristics as those which the intended scanning environment will use. If, as ~~may~~can happen, an application specification does not specify the light source, a judgment must be made in order to determine the most probable light source for reading, to enable valid measurements to be made and to be sure that the results will be properly indicative of likely scanning performance in the application.

It should be noted that for maximum correlation, it is not only the light source (including any filters that modify its spectral distribution) that must be taken into account, but also the spectral sensitivity of the sensor, since reflectance at a given wavelength is a function of the product of the intensity of the light emitted and the sensitivity of the sensor. However, for the purposes of this Annex, the sensor sensitivity is ignored.

In general, the sensor should be nominally linear and not enhanced in any way, as described in 5.6.3. Specific care shall be taken if any filters are used since they can modify the raw image significantly and produce results that are not reproducible. Filters, including polarization, low pass, high pass, or notch filters, ~~etc.~~ should only be used ~~in accordance with~~according to an application specification which specifies the use of such filters for an application specific purpose, ~~and in~~. In any case in which ~~the~~ use of a filter significantly affects a measurement, ~~then~~ the use of the filter should be reported along with the grade.

C.1.2 Light sources

Light sources for bar code scanning applications normally fall into two areas:

- narrow band illumination in either the visible or the infra-red spectrum, or
- broadband illumination covering a large part of the visible spectrum, sometimes referred to as "white light" although it ~~may~~can have a bias to a colour; ~~a~~ very few specialised applications may call for light sources of unusual characteristics such as ultra-violet for fluorescent symbols.

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Multi-row bar code scanning almost always uses narrow band visible light, with light sources with a peak wavelength in the red part of the spectrum, between 620 and 700 nm. Infra-red scanning uses sources with peak wavelengths between 720 nm and 940 nm.

Two-dimensional matrix symbols are scanned under a variety of illumination conditions, with the most common being white light and, in a number of hand-held reading devices, the same visible red area of the spectrum as for linear and multi-row bar code symbols.

The most common light sources, but not a complete list, that may be used for these purposes are:

a) ~~Narrow~~narrow band:

- 1) ~~Light~~light-emitting diode (near-monochromatic, at numerous visible and infra-red peak wavelengths), and
- 2) ~~Solid~~solid-state laser diode (most usually 650 nm and 670 nm) (multi-row bar code symbols only);

b) ~~Broadband~~broadband:

- 1) ~~Incandescent~~incandescent lamp (nominally white light with a colour temperature in the range 2800 K to 3200 K), and
- 2) ~~Light~~light-emitting diode (LED) with nominally “white light” characteristics that emit different intensities for red, green and blue components. ~~Their, their~~ actual spectral distribution ~~may~~can show a number of peaks ~~e.g., for example,~~ in the blue and yellow or orange regions ~~with a colour temperature in the range of 2000 K to 7000 K) depending on the specific LED type.~~

NOTE 1 A formal grade including a light specifier of /W(7000 K) is an example of white light-emitting diode illumination.

- 3) ~~Halogen~~halogen lamps (nominally white light with a colour temperature in the region of 2800 K to 3200 K), and
- 4) ~~Gas~~gas discharge lamps (light of various characteristics);

NOTE 2 Broadband illumination is often the ambient light in a scanning environment.

The key characteristics of these are as follows.

A light-emitting diode is a low-power solid-state component most frequently found as the light source in a light pen (wand) or CCD (charge coupled device) scanner. Operating wavelengths in the visible spectrum may be from 620 to 680 nm; most commonly either 633/640 or about 660 nm. In the infra-red spectrum, 880 to 940 nm is the most common range of wavelengths. In the UV (ultraviolet) spectrum, 365 ~~nm to~~ 395 nm is the most common range of wavelengths.

A laser diode is also a low power solid-state component emitting highly monochromatic coherent light. Typical wavelengths in the visible spectrum used by these, at the date of publication of this document, is 650 nm. In the infra-red spectrum, 780 nm is common. They are frequently found in hand-held (laser) scanning equipment and a number of fixed scanners.

Broadband light sources are mainly found in systems using two-dimensional imaging and image processing technology rather than scanning techniques.

Incandescent lamps have a power distribution covering much of the visible spectrum and well into the near infra-red spectrum; their optical characteristics are more easily defined in colour temperature terms rather than in those of peak wavelength, because of the wide bandwidth and relative absence of clearly-defined peaks in the power distribution. These broadband power distribution characteristics mean that the symbol contrast values obtained from symbols may vary with different colour temperatures to a significantly lesser

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extent than values obtained with light sources whose power distributions peak sharply with narrow bandwidth.

Halogen lamps (also known, more correctly, as tungsten halogen lamps) are a development of incandescent lamps with a higher colour temperature and a smooth power distribution curve across the spectrum, extending well into the near infra-red.

Fluorescent light sources also produce nominally white light and have broadband power distribution characteristics, which, in comparison with those of an incandescent source, tend more towards the bluer region of the visible spectrum, often with a significant ultra-violet component, and a number of peaks in their spectral power distribution. Typical colour temperatures for such lighting are in the region of 3200 K to 5500 K. The physical structure of a fluorescent lamp is that of a tube which can be formed into various shapes, and an annular shape concentric with the optical axis of a reading device provides very satisfactory uniform diffuse illumination.

Light emitting diodes with nominally "white light" characteristics emit "cool" white light and may have a nominal colour temperature in the region of 7000 000 K. Their actual spectral distribution may can show a number of peaks e.g., for example, in the blue, and yellow or orange regions.

Gas discharge lamps tend to have spectral distributions with multiple sharp peaks at wavelengths depending on the precise mixture of gases used. For example, sodium vapour emits light with a well-defined peak at around 580 nm (yellow-orange) and mercury vapour emits a green-blue light at around 520 nm.

The use of filters to modify the spectral distribution of the illumination system is common. For example, when used in conjunction with a Wratten 26 filter, the light characteristics of a 2856 2 856 K lamp approximate to those of a 620-633 nm source. The use of infra-red and/or ultra-violet absorbing filters is also common in scanning systems. It is possible to alter the apparent colour temperature of a source by the use of using filters.

NOTE 3 Light source types, wavelengths and colour temperatures stated above are indicative and may can change as the technology evolves. Helium-neon lasers for example are not common anymore and has been removed from the above description in this revision edition of this document.

C.1.3 Effect of variations in wavelength

The reflectance of a substrate or bar code symbol element varies with the wavelength of the incident light. A black, blue, or green printed area will tend tends to absorb visible red light strongly (and appear therefore of low reflectance), whereas a white, red or orange area will reflect reflects most of such incident light. In the infra-red spectrum, the apparent colour of the element does not correlate at all with reflectance; it is the nature of the pigmentation used (for example e.g. the proportion of carbon content) which governs reflectance. Taking reflectance measured at 633 nm as a reference, when measured at 660 or 680 nm the results may can differ significantly, and sufficiently to cause the symbol grade to change by one or two units, or even more in the case of bars printed on some thermal papers.

In the case of broadband illumination, however, the presence of light at multiple wavelengths in the spectral power distribution of the light means that reflectance values of black inks measured under white light from various sources will does not differ significantly; there may however, there can be some variation (an increase in reflectance) in the case of dye-based black inks where the illumination has a significant infra-red component. With coloured pigments, there will be is greater variations. Interposing a filter in the light path will introduce introduces a more peaked spectral distribution and the spectral response curve of the reader

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will require to be more closely matched to that of the light source. It is common for the optical system to include both infra-red and ultra-violet absorbing filters.

C.1.4 Considerations affecting selection of broadband light sources

Broadband light sources, by definition, emit light over a band of wavelengths without a clearly defined sharp peak. Nonetheless, the intensity of light emitted at different wavelengths will vary. In particular, the light of a colour temperature in the region of 3000 K is described as "warm" light and the spectral distribution of this light shows higher intensity of emission towards the red (and infra-red) region of the spectrum, whereas light with a higher colour temperature in the region of 6500 K is described as "cool" light and its spectral distribution is biased to the blue-violet region of the spectrum, extending into the ultra-violet. Light with a higher colour temperature will yield higher reflectance values on blue pigments than light with a lower colour temperature. The converse is true for red pigments.

It is possible to modify the apparent colour temperature of a light source by the interposition of an appropriate filter.

It maycan also be possible to approximate the characteristics of different broadband light sources with sufficient precision for bar code symbol quality assessment purposes by combining reflectance measurements at three narrow band wavelengths across the visible spectrum, e.g. in the red, green and blue regions (assuming that the ultra-violet and infra-red regions have been cut off by the use of appropriate absorbing filters); the results can then be modified to match the spectral response characteristics in the application by applying an appropriate correction factor at each wavelength.

C.2 Selection of aperture

The choice of aperture size is very important, and it must be specified in accordance with 5.4 for matrix symbol grading and ISO/IEC 15416 for stacked symbol grading, in order for symbol grades to be measured consistently. It is the responsibility of an application specification to define an appropriate measuring aperture to be used. As required by 5.5, the aperture size must be reported together with the grade and illumination in order to identify the conditions under which the measurement was made.

NOTE 1 The term "aperture" does not refer to the opening of a lens. ~~It refers but~~ to the convolution of the raw image.

The size of the measuring aperture affects whether voids in the symbol will be "filled in" during the verification process. Therefore, the measuring aperture must be selected with reference to the range of nominal module size and expected scanning environment. An aperture that is too small willis not able to fill in unintentional voids, or gaps between elements of a direct marked symbol, and that would lead to low grades or undecodable symbols. On the other hand, a measuring aperture that is too large will blueblurs individual modules, resulting in low modulation, and maycan prevent the symbol from being decoded.

An aperture size in the range of 50 % to 80 % of the minimum allowed module size is a typical choice for an application specification. Importantly however, an application specification that allows a range of nominal module sizes (for example e.g. a range of 0,25 mm through 0,40 mm) should specify a single aperture size to be used in all cases or should specify more than one acceptance criteria, each with a different aperture size. If more than one acceptance criteria is allowed in an application, scanning all of the acceptable symbols will bebecomes more challenging to practical readers with fixed scanning parameters.

NOTE 2 In applications in which a single aperture size does not adequately predict scanning performance, the use of ISO/IEC 29158 mayshould be preferred over used instead of this document.

The reason that more than one aperture size maycan be desirable in an application is that the relatively small measuring aperture required to read the symbols with the smallest module size will limitlimits the size of the largest acceptable spots and voids. But, if tooan overly large an aperture wereis used, the modulation for the smallest module size would bebecomes inadequate. In general, the larger the aperture, the larger the acceptable size of spots and voids. Conversely, the smaller the aperture, the smaller the acceptable module size that can be read. Therefore, a successful application specification must select a measuring aperture that

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~~will predict~~**predicts** the readability of both the largest and smallest module size symbols. If an application truly contains symbols with a large range of X dimensions, ~~it may be desirable to use~~ different aperture sizes ~~should be used~~ in order to optimize the performance for all the symbols by using different aperture sizes and thereby offering tolerance of spots and voids in all the symbols to a larger extent than would be possible if using an aperture that was appropriate of the smallest X dimensions.

The choice of measuring aperture(s) specified ~~may can~~ be influenced by the scanning equipment that is expected to be prevalent in the application-scanning environment. Conversely, the ~~choice of the~~ scanning equipment ~~may can~~ also be influenced by the specification of the measuring aperture(s). In both cases, however, a “match” between ~~the~~ verification technique and ~~the~~ scanning environment is made in order to produce a high correlation between grade level and scanning performance.

Table C.1 may be used by application specification writers as a guide. When the measuring aperture diameter is not specified in the application specification, Table C.1 should be used as a guide when applying this document for process control purposes.

NOTE 3 An application specification ~~may can~~ specify a range of X dimensions that differs from those listed in Table C.1 and can specify an aperture size that differs from the recommendation of Table C.1.

Table C.1 — Guideline for diameter of measuring aperture

X Dimension(dimension mm)	Aperture diameter (mm)	Reference number	
$0,037 \leq X < 0,100$ (1,5 mil- to 4 mil)	0,025	01	
$0,100 \leq X < 0,150$ (4 mil- to 6 mil)	0,050	02	
$0,150 \leq X < 0,190$ (6 mil- to 7,5 mil)	0,075	03	
$0,190 \leq X < 0,250$ (7,5 mil- to 10 mil)	0,125	05	
$0,250 \leq X < 0,500$ (10 mil- to 20,0 mil)	0,200	08	
$0,500 \leq X < 0,750$ (20 mil- to 30 mil)	0,400	16	
$0,750 \leq X$ (equal to or greater than 30 mil-))	0,500	20	
NOTE The aperture reference number approximates the measuring aperture diameter in thousandths (“mil”) of an inch; this reference number is used for consistency with the ANSI X3.182 and ISO/IEC 15416. When the aperture is less than 0,025, its actual size is reported using actual dimensional units, so that a reference number of 00 is not used.			

C.3 Selection of lighting angle

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Whereas the default 45° illumination angle is well suited to printed symbols and those marked on even surfaces without localised points of specular reflection, i.e. those from which diffuse reflection does not vary sharply with the angle of incidence or collection of the light, many "direct marked" symbols require to have the angle of incidence adjusted to optimise reading performance. The spectral characteristics of the illumination for use with engraved or similar symbols may therefore be less important than the respective angles of incidence and collection of the light on the symbol. The light source needs to be positioned in such a way that the apparent contrast as seen by the image capture device is related to the process used in the application to read the symbols.

Depending on the nature of the surface and the marking technique used, the spectral characteristics of the light source ~~may~~ also have an influence on its contrast. Prediction of scanner performance will be enhanced when the application specification requires verification using the same source as that used for reading the symbols. The modified version of the methodology defined in ISO/IEC 29158 ~~may~~ provide a better approach to select appropriate lighting angles for direct part marks.

C.4 Selection of minimum acceptable grade

The specification of the minimum acceptable grade in an application specification should be based on consideration of the trade-off between the possibly increased cost of producing higher-grade symbols and the improved scanning performance to be obtained by the use of such symbols, together with the data integrity requirements of the application.

A requirement for higher grade symbols may restrict the choice of the following available to the producer of the symbols:

- ~~Inks~~ (or other marking media) and substrates on which to mark the symbol (e.g. to ensure a high level of symbol contrast, a substrate with high reflectance and/or an ink with low reflectance under the specified illumination are needed, imposing limits on the choice of colour available);
- ~~Marking~~ technology (e.g. those in which the placement of printer dots is less well-controlled may be excluded).

It ~~may~~ also require slower production rates or higher levels of quality control, or lead to higher rejection rates, all of which contribute to higher unit costs.

On the other hand, the receiver of the symbols will benefit from improved read rates, or may have a greater choice of reading technology.

If a low symbol grade is specified, the receiver of the symbols ~~may~~ incur additional costs in:

- ~~Installing~~ higher quality reading equipment;
- ~~Accepting~~ a lower read rate;
- ~~Reprocessing~~ of symbols that fail to scan.

Many applications specify a minimum grade of 1,5 which offers them a fair balance between the cost of production and ~~the~~ reading performance under the conditions of their application.

The more critical it is that a high read rate be achieved, on grounds not only of cost but also data integrity, the higher the grade that needs to be specified.

C.5 Effect of ~~Symbol Rotation~~ symbol rotation during verification

The first edition ~~publication~~ of this document called for an overall grade to be computed by averaging five individual scan results, taken with the symbol rotated in five different orientations. However, this

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requirement has been removed and this document now calls for only one scan to be used for grading. The five-scan rotation requirement was originally required for two reasons:

— to account for symbols that do not exhibit uniform diffuse reflection, and

— to “average out” any effective changes in measurement resolution by verification devices.

The first reason is better addressed through a modified version of the methodology defined in this document in ISO/IEC 29158. The second reason is addressed ~~by the publication of~~ ISO/IEC 15426-2 which requires the minimum effective resolution defined in 5.6.3.

The removal of the five-rotation requirement greatly simplifies the verification procedure for most symbols (whose characteristics do not vary with orientation) and facilitates practical quality control regimes.

Annex D
(informative)

Substrate characteristics

D.1 Introduction

In certain circumstances, for example, the design and production of printed packaging materials incorporating bar code symbols or the production of symbols directly marked on to a surface, it ~~may~~ can be necessary or desirable to assess the acceptability of ~~either~~ substrates ~~and/or~~ ink colours, ~~or both~~, for a given bar code application, before a bar code symbol is available ~~which could be tested in accordance with~~ ~~for testing according to~~ this document. Reference should be made to ISO/IEC TR 19782 for additional guidance on the effects of gloss and low substrate opacity on the reading and verification of these symbols.

D.2 Substrate opacity

The methodology of ~~the~~ ~~this~~ document requires that the symbol shall be graded according to the parameters in ~~clause~~ ~~Clause~~ 6 ~~(for multi-row symbols)~~ or ~~Clause~~ 7 ~~(for matrix symbols)~~, when measured in its final configuration, e.g. final filled package.

If it is not possible to measure the symbol in this configuration, then the effects of show-through of high-contrast interfering patterns ~~may~~ can be ignored if when measured as follows the substrate opacity is 0,85 or greater. If the opacity is less than 0,85, the symbol should be measured when backed by a uniform dark surface the reflectance of which is not more than 5 ~~per cent~~ %.

The opacity of the substrate shall be calculated as follows:

$$P = R_2 / R_1$$

~~Where~~

~~where~~

- ~~P~~ ~~Opacity is the opacity:~~
- ~~- R_1 is the reflectance of a sample sheet of the substrate backed with a white surface the reflectance of which is 89 % or greater;~~
 - ~~- R_2 is the reflectance of the same sample sheet backed with a black surface the reflectance of which is not more than 5 %.~~
-
- ~~- R_1 Reflectance of a sample sheet of the substrate backed with a white surface the reflectance of which is 89 percent or greater.~~
 - ~~- R_2 Reflectance of the same sample sheet backed with a black surface the reflectance of which is not more than 5 percent.~~

D.3 Gloss

The reference illumination conditions specified for the measurement of reflectance should enable the maximum rejection of specular reflection while giving a representative assessment of the diffuse reflectance of the symbol and substrate. Highly glossy materials and those whose diffuse reflectance characteristics vary

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with ~~either~~ the angle of incident ~~and/or~~ collected light ~~—, or both —~~ as ~~may~~**can** be the case with many materials on which symbols are directly marked ~~—may— can~~ yield grades differing from those obtained by the use of the reference optical arrangement with illumination at 45° ~~—and for—~~. For this reason ~~sub-clause~~, 5.4 provides alternative angles of illumination to enable apparent symbol contrast to be maximised.

D.4 Over-~~laminat~~**lamination**

~~The methodology of this document requires that the symbol shall be graded when measured in its final configuration. A symbol intended to be covered with a protective lamination should be graded according to the parameters in clause 6 (multi-row symbols) or 7 (matrix symbols) shall be graded when measured with the laminate in place. The Using a laminate with thickness of the laminate, including its adhesive should be, that is as small as possible in order tends~~ to minimise its effects on the reading performance of the symbol.

D.5 Static reflectance measurements

D.5.1 General

In some cases, it ~~may be~~**is** desirable to carry out static reflectance measurements of samples of the substrate on which a bar code is to be printed and on colour patches or ink samples which replicate the colour in which the bar code will be printed. The following guidelines provide a means which, if it is followed, will predict as closely as is generally possible the results which will be obtained when the symbol is scanned dynamically.

Static reflectance measurements should be made with the wavelength of light, aperture size and optical arrangement which relate to the application and which are specified in accordance with this document. Where reflectance measurement equipment meeting the requirements of this Annex is not available, optical density measurements may be made using a standard densitometer with an appropriate light source and converted to reflectance values; density (D) and reflectance (R) are related as follows:

$$R = 100 / 10^D$$

NOTE It is impossible to predict to a high degree of accuracy the symbol contrast and, in particular, the edge contrast which ~~will be~~**is** achieved in the printed symbol. It is therefore appropriate to allow some safety margin above the minimum values for specified grades.

D.5.2 Prediction of ~~Symbol Contrast (SC)~~**symbol contrast**

The prediction of symbol contrast requires that measurements of reflectance be made on samples which simulate the highest ($R_{\max}$) and lowest reflectance ($R_{\min}$) areas which will be present in the finished symbol.

It is probable that in most bar code symbols $R_{\max}$ will be found in the quiet zone of the symbol; therefore, to simulate the conditions found in the quiet zone, $R_{\max}$ should be measured in the centre of a sample area, at least 10X in diameter, of the material on which the symbol is to be printed.

It is probable that in most bar code symbols $R_{\min}$ will be found in the widest bars of the symbol, or areas with a number of contiguous dark modules; therefore to simulate the conditions most likely to yield a value of $R_{\min}$ consistent with that which would be found in practice, reflectance should be measured in the centre of a strip of material 2X to 3X wide and which matches the colour in which the dark elements are to be printed.

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A predicted value of SC can then be calculated:

$$\Delta R'_{SC} = R_{\max} - R_{\min}$$

For materials which do not satisfy the tests for opacity, which are detailed in ~~Annex~~ Clause D.2, the measurements which are made for the purpose of predicting SC should be made with the test samples backed by a uniform dark surface, the reflectance of which is not more than 5 %. The same measurements should then be made with the test samples backed by a uniform surface the reflectance of which is not less than 89 %. The calculated value of static SC shall be equal to or greater than the minimum value for the grade selected for the application, for tests on both the dark and light backgrounds.

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Annex E (informative)

Parameter grade overlay applied to two-dimensional symbologies

This Annex describes the technique used in this document to derive a final grade for a parameter from a set of notional grades determined for a set of grade levels for the parameter. In this revision to this document, the number of grade levels is 41, corresponding to a grading scale of 4.0 to 0 with steps of 0,1.

The technique computes a notional grade for a parameter for each grade level by assuming that only modules or codewords that meet or exceed that grade level for that parameter are actually readable. The modules or codewords which are readable then result in a grade for that parameter according to the rules for that parameter (whether based on unused error correction or fixed pattern damage).

If one considers what the performance would be for a scanner that ~~could~~can only read codewords or modules above a particular parameter grade level, it is clear what will happen – only codewords or modules at or above that grade level may~~can~~ be counted towards the readability calculation for the symbol at that grade level.

For example, if codewords or modules with a grade of 2 must be counted before a grade of 3 on unused error correction or fixed pattern damage may~~can~~ be obtained, then the symbol must be a grade 2.

Furthermore, if codewords graded 3 or better can only result in an unused error correction or fixed pattern damage level of 2, the symbol must also be a grade 2.

However, the readability of a symbol must take into account the readability of codewords or modules at each grade level and the ability of the symbol to be read using error correction or allowing for some fixed pattern damage and the resulting grade should be the highest of these two possible outcomes.

The following procedure can therefore be established:

- Count the number of codewords in each grade level, including higher grade levels, assume that all remaining codewords are erasures (multi-row symbols) or errors (matrix symbols) and determine ~~at the~~ unused error correction or fixed pattern damage grade.
- For each grade level, take the lower of the grade level and the associated unused error correction or fixed pattern damage parameter grade.
- Select the highest of the values from step b) as the symbol grade for that parameter.

This ensures that a scanner will have performance associated with the assigned grade because the scanner's ability to read codewords or modules of the assigned grade or higher will bring it within the error correction or fixed pattern damage capacity of the assigned grade level or better.

This method provides a way of accounting for imperfections in symbols which are designed to tolerate imperfections. In fact, it favours symbols with more error correction capacity, which certainly does make a symbol easier to read. It also reconciles the print quality measurement method of linear symbols with that of 2D symbols. In a sense the linear approach, which takes the worst case, is the trivial extension of the above rule in the case of no error correction. In this case, the codeword with lowest grade is always needed to get

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anything other than a 0 for “unused error correction”. If this value happens to be a 1, then the symbol must be a 1, even if all other codewords had a quality of 4.

NOTE The notional ~~Unused Error Correction~~unused error correction or ~~Fixed Pattern Damage~~fixed pattern damage grade used in this calculation is not related to, and does not affect, the unused error correction or ~~Fixed Pattern Damage~~fixed pattern damage grade for the symbol as calculated according to 6.2.4, 7.5.6, or 7.5.9.

Annex F
(informative)

Explanation of ~~some~~the main changes in this ~~revision~~edition of this document

F.1 Continuous ~~Grading~~grading

A significant change that is introduced in this revision is the use of “continuous grading”, wherein grades are rounded to the next lowest 0,1 (tenth) of a grade level. This results in more granular grading by interpolating grades between the integer grade boundaries that were defined in earlier versions. This reduces troubling fluctuations in grades when small changes in measurements cause a grade to transition between integer grade levels. Grades are to be rounded down to the nearest tenth so that the boundary between whole numbers remains unchanged from the previous ~~revision~~edition of this document. See ISO/IEC TR 24244 for a further discussion on continuous grading.

~~Example—~~

EXAMPLE—In the previous ~~revision~~edition of this document, a value of ~~Symbol Contrast~~symbol contrast in the range of greater than or equal to 0,55 and less than 0,70 was graded as a 3. According to this new method of “continuous grading”, a symbol contrast value of 0,699 would be graded 3,9. Note that the value has almost attained the minimum required for a grade of 4,0, but it is graded 3,9 because it is still less than 0,70 which is required to attain a grade of 4,0.

F.2 Threshold

The previous ~~revision~~edition of this document computed the global threshold used to binarise the reference grey-scale image of 2D matrix symbols as the midpoint between the highest and lowest value of reflectivity found in the reference grey scale image. In this ~~revision~~edition of this document, the threshold is calculated using the method defined in Annex A and is based on the histogram of reflectance values in the reference grey scale image. This is less sensitive to the extremes of reflectance at the “tails” of the histogram, but instead is calculated to produce the minimum variance in the two sub-histograms produced when separated by the threshold. This threshold is not referred to as a ~~Global Threshold~~global threshold, but simply as “~~Threshold~~”. ~~The~~threshold. However, the use of ~~Global Threshold~~global threshold for stacked multi-row is not changed in this ~~revision~~edition of this document.

F.3 Redefined $R_{\max}$ and $R_{\min}$ for 2D matrix symbol grading

This ~~revision~~edition of this document modifies the computation of ~~$R_{\max}$~~ $R_{\max}$ and ~~$R_{\min}$~~ $R_{\min}$ for 2D ~~Matrix~~matrix symbols. The previous ~~revision~~previous edition of this document defined ~~$R_{\max}$~~ $R_{\max}$ as the maximum reflectance within the symbol area and ~~$R_{\min}$~~ $R_{\min}$ as the minimum reflectance within the symbol area. This means that the very brightest and darkest pixel would determine ~~Symbol~~the symbol contrast, and have a significant impact on the calculation of ~~Modulation~~modulation (these values figure in the denominator) for all modules. ~~In the revision~~In this edition of this document, ~~$R_{\max}$~~ $R_{\max}$ is defined as the average of the brightest 1 % of the pixels and ~~$R_{\min}$~~ $R_{\min}$ is defined as the average of the darkest 1 % of the pixels. In this way,

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the grading of ~~Symbol Contrast~~symbol contrast and ~~Modulation will be~~modulation are less affected by small areas of extreme reflectance. This affects ~~Symbol Contrast~~symbol contrast and ~~Modulation~~modulation grading, and ~~will tend~~tends to:

- increase ~~Modulation~~modulation grades,
- decrease ~~Symbol Contrast~~symbol contrast grades, and
- increase the stability of ~~Modulation~~modulation and ~~Symbol Contrast~~symbol contrast grades.

This change to $R_{\max}$ and $R_{\min}$ is not applied to ~~clause~~Clause 6, where $R_{\min}$ and $R_{\max}$ are used as the basis for computing the ~~Global Threshold~~global threshold.

F.4 Print ~~Growth~~growth

This ~~revision~~edition of this document adds a generic grading method for ~~Print Growth~~print growth which ~~may can~~ be required by ~~Symbolology Specification~~symbolology specifications (see F.5.2). ~~This section~~ — this clause explains why this has been added.

High print growth distorts the nominal characteristics of a symbol, such as the nominal 1:1 ratio of the elements within clock tracks, or the nominal ratio between the elements of finder patterns. Furthermore, it reduces the size of some elements (typically space elements) such that the optical resolution required for reading is higher than normally required for symbols with the same X-dimension.

NOTE Print ~~Growth~~print growth refers to the dark elements of symbols in the normal, or positive, polarity and to the enlargement of the light elements in a symbol of reverse polarity. Where these elements are reduced in size, ~~Print Growth~~print growth is negative.

One significant issue is effective resolution which is represented by the aperture size in a verifier. However, the measuring aperture used in verification is not directly reproduced by reading equipment, and this may lead to applications in which symbol grade is not a good predictor of scanning performance. The reduction in element size caused by ~~Print Growth~~print growth would affect ~~Modulation~~modulation, but not in cases where the aperture size is small relative to the nominal X-dimension. Importantly, in applications in which a range of X-dimensions is allowed, and the ~~Aperture Size~~aperture size is specified as a size that is smaller than the smallest allowed X-dimension, the Aperture may indeed be significantly smaller than the elements of a symbol with the largest allowed X-dimension, even with high ~~Print Growth~~print growth.

High print growth in a symbol ~~may or may not can~~ affect readability or reading performance in particular reading environments (largely depending upon optical resolution and decoding algorithm). Verification of symbol quality is intended to both predict readability and also to offer process control feedback in symbol production. However, verification algorithms (application of synthetic aperture and the application of the reference decode algorithm) is not a perfect model of real-world reading systems. In particular, the actual effective resolution of reading systems changes throughout the depth of focus, or reading range. Real world reading systems are designed to balance high read rate, high speed, cost, and ability to read symbols of varying quality. The maintenance of symbol characteristics close to their nominal values is of high value in achieving these goals. Since high ~~Print Growth~~print growth potentially affects reading system performance and ~~may is~~ not ~~be always~~ captured in the grading in any other way, ~~Print Growth~~print growth as a graded parameter ~~may can~~ be the only way to capture this aspect of symbol quality.

Figure ~~HE~~.1 illustrates a symbol with ~~Print Growth~~print growth of approximately 50 % of the X-dimension. This means that the 1 module dark elements are three times as wide (or tall) as 1-module light elements. Since these elements are nominally supposed to have a ratio of 1:1, this 3:1 ratio represents a large deviation from the intended nominal characteristics of the symbol structure.

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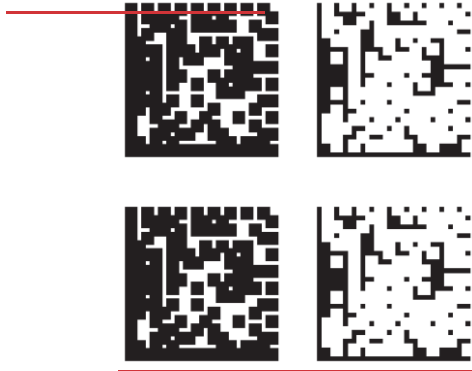


Figure HF.1 — Symbol exhibiting very high print growth/and reduction

The ~~symbol~~symbol illustrated in ~~figure H~~Figure F.1 has a nominal X-dimension of 0,5 mm (not drawn at scale) would obtain a grade of 4.0 when verified with an 08 aperture, prior to the addition of the ~~Print Growth~~print growth grade. The aperture of 08 corresponds to the aperture size that would typically be specified in an application in which the allowed range of X-dimensions was 0,25 mm to 0,5 mm. When the ~~Print Growth~~print growth grade is used however, the grade is 1,0.

F.5 Other changes

F.5.1 Improved guidance to application specifications

This ~~revision~~edition of this document contains added guidance regarding the requirement for application specifications to define several parameters necessary in applying the method defined in this document (see 5.1, 5.4, 5.6.3, 6.2.2, 7.3.1, and Annex C). There is also added guidance saying that the application specification ~~may can~~ define more than one acceptance criteria, which enables the application to permit more than one aperture size to be used in order to obtain an acceptable grade in cases where this is appropriate for an application.

F.5.2 Guidance to symbology standards

This ~~version~~edition of this document adds the suggestion that symbology specifications ~~may wish to can~~ incorporate a graded parameter for ~~Print Growth~~print growth (see 7.5.11.2).

F.5.3 ~~Removed Extreme Reflectance~~Deleted extreme reflectance check

The previous ~~revision~~edition of this document contained a provision for grades to be indicated with an asterisk, to indicate the presence of extreme reflectance in the area within or near the symbol. This has been removed in this ~~revision~~previous edition of this document (see 5.4).

F.5.4 Modulation and ~~Reflectance Margin~~reflectance margin merged into one parameter

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The previous ~~revision~~ ~~edition of this document~~ defined both ~~Modulation~~ ~~modulation~~ which was graded without regard to “errors” and ~~Reflectance Margin~~ ~~reflectance margin~~ which was graded with sensitivity to “errors”. These were combined into one parameter, now called ~~Modulation~~ ~~modulation~~, which regards errors. Therefore, the parameter which used to be called ~~Reflectance Margin~~ ~~reflectance margin~~, is now called ~~Modulation~~ ~~modulation~~ and the old ~~Modulation~~ ~~modulation~~ parameter is removed (see 7.5.5.1).

F.5.5 Minor editorial changes

Minor editorial improvements were made throughout this document and are intended to improve the clarity and answer some of the questions which experience has shown ~~may~~ ~~can~~ arise out of reading the previous edition.

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¹ Specifications published by AIM Global - viz. AIM International Technical Specifications (ITS) and Uniform Symbology Specifications (USS) - are obtainable from AIM Inc., 634 Alpha Drive, Pittsburgh, PA 15238, USA

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Composite ~~Symbology~~symbology

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~~NOTE 1 Specifications published by AIM Global — viz. AIM International Technical Specifications (ITS) and Uniform Symbology Specifications (USS) — are obtainable from AIM Inc., 634 Alpha Drive, Pittsburgh, PA 15239, USA~~

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